Journal of Energy Storage 99 (2024) 113355

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Research papers

Off-design operation and performance of pumped thermal energy storage

Joshua McTigue *, Ty Neises

National Renewable Energy Laboratory, 15013 Denver West Parkway, Golden, CO 80401, USA

ARTICLE INFO

Keywords:
Carnot battery
Pumped thermal energy storage
Molten salts
Inventory control
Off-design

ABSTRACT

In this article, we describe off-design models and control strategies for a Pumped Thermal Energy Storage (PTES) system that uses liquid thermal energy storage: specifically molten salt for hot storage and methanol for cold storage. Off-design conditions arise when load-following, or due to variations in storage tank temperatures or ambient temperatures. We propose a control strategy that uses inventory control to manage the mass flow rate in the thermodynamic cycles, which facilitates load following. We also propose a control strategy for the storage fluid mass flow rates, which are varied to ensure the molten salt is maintained at its design temperature. This maximizes efficiency and minimizes problems with salt freezing or degradation. The cold storage fluid mass flow rate is varied so that the cold tanks have the same state-of-charge as the hot tanks. This leads to variations in cold fluid temperature, but these variations are shown to be acceptably small (e.g. 7.5 % increase), and this control method is shown to be simpler and more efficient than an alternative strategy where tanks become unbalanced. The ambient temperature and storage tank temperatures are moved ±50 °C from the design values and the impact on power, duration, and tank temperatures is quantified. Results demonstrate that the proposed control strategy is stable and self-correcting – that is, storage temperatures converge on stable values after two-to-three charge-discharge cycles. When inputs return to design values, the system returns to its design point after two charge-discharge cycles. We also demonstrate that inventory control enables delivery of the target power output even when off-design conditions exist that would normally reduce the power output.

1. Introduction

Most of the recently deployed energy storage in the USA has discharge durations of 4 h or less – partly as a result of current capacity market structures [1]. These systems are well-suited for managing peaks in summer net demand (demand minus renewable generation) which tend to last less than 4 h. However, in several regions of the USA, net peaks are greater during the winter than the summer, and these peaks also tend to last longer [1]. Meanwhile in the UK, the prominence of wind generation has led to longer periods of renewable surplus and, correspondingly, longer periods of renewable deficit [2]. Net zero emission roadmaps suggest that there will be a significant increase in variable renewable electricity generation by 2050 [3]. Thus, there is a growing opportunity for Long-Duration Energy Storage (LDES) systems to provide value to the grid by delivering system flexibility and reliability. The California Public Utilities Commission has recently proposed procurement of 1 GW of >12 h storage and 1 GW of multi-day energy storage [4] which demonstrates the need for LDES. In this article, we investigate the control and performance of an LDES system for a range of off-design conditions and provide a framework for understanding how these systems behave in real-world scenarios.

LDES discharge power for 6–10 h or more and are typically characterized by low marginal costs of energy storage capacity [5], which can be achieved by using, for example, thermal energy storage (TES) media, hydrogen, or compressed air. A Carnot Battery is one such LDES system that can use a variety of TES materials, such as water, rocks, molten salts, or sand [6]. These materials enable Carnot Batteries to have flexibility in site location and deliver large energy and power capacities without requiring critical or rare materials. Carnot Batteries also have the potential for "sector-coupling" – that is, integrating sources of electricity and thermal energy for storage, and later delivering electricity, and hot and cold energy to maximize the efficiency and value of the system [7].

Pumped Thermal Energy Storage (PTES) is a subset of Carnot Batteries which are characterized by using heat pumps and heat engines to convert between electricity and thermal energy. PTES uses a heat pump to extract thermal energy from a source reservoir and deliver it at a higher temperature to a sink reservoir. PTES therefore creates both hot and cold thermal energy storage. The stored thermal energy is later

* Corresponding author.
E-mail address: jmctigue_nrel@outlook.com (J. McTigue).

https://doi.org/10.1016/j.est.2024.113355
Received 30 May 2024; Received in revised form 6 August 2024; Accepted 11 August 2024
Available online 19 August 2024

Nomenclature

Greek letters

γ	Ratio of specific heats c_p/c_v
ε	Heat exchanger effectiveness
η_p	Polytropic efficiency
η_{HE}	Heat engine efficiency
η_{RT}	Round-trip efficiency
ρ	Density, kg/m^3
τ	Duration of charge or discharge, h

Roman letters

A	Heat exchanger heat transfer area, m^2
h	Enthalpy, J/kg.K
f_p	Pressure loss factor
$\dot{m}$	Mass flow rate, kg/s
$\dot{m}_c$	Mass flow coefficient
$\dot{m}_r$	Relative mass flow coefficient, $\dot{m}_c/\dot{m}_{c,o}$
p	Pressure, bar
$\dot{Q}$	Heat, W
T	Temperature, K
$\dot{W}$	Power, W

Subscripts

amb	Ambient conditions
chg, dis	Charge, discharge
comp, exp	Compressor, expander
h, c	Hot, cold
o	Design point conditions
sink, source	Sink, source
SF	Storage fluid
WF	Working fluid

Abbreviations

COP	Coefficient of Performance
LDES	Long-Duration Energy Storage
PTES	Pumped Thermal Energy Storage
SOC	State-of-Charge
TES	Thermal Energy Storage

converted back into electricity with a heat engine that operates between the two thermal reservoirs.

A recent report for the International Energy Agency (IEA) Technology Collaboration Programme on Energy Storage examined the potential of Carnot Batteries and compiled details of modelling efforts on those systems [8]. The report noted that system-level design could benefit from an improved understanding of the effects of off-design operation. The report also suggested that there is a lack of understanding of Carnot Battery operating strategies and dynamic behavior, and that developing control strategies would clarify the services that Carnot Batteries could provide currently and in the future.

Off-design operation of PTES will occur for several reasons. For PTES to provide load following services the system may be required to operate at part-load. Off-design operation also occurs when storage tank temperatures or ambient temperatures deviate from their design values. This can have an instantaneous impact on the efficiency and power input/output and also affects the performance of future cycles: delivering off-design temperatures to the storage tanks in the current cycle will then cause subsequent cycles to operate at off-design conditions. Similarly, off-design operation can cause storage tanks to become 'unbalanced' (i.e. the state-of-charge of hot and cold tanks differs) which has long-term consequences. As a result, the actual round-trip efficiency over the operational lifetime may be quite different to the design-point round-trip efficiency which is the performance metric that is traditionally used to evaluate these systems. Control strategies should manage off-design conditions in a robust manner and must also ensure that operational limits are not exceeded – storage fluids must not be operated outside of their operational temperature range to avoid freezing, boiling, or decomposition, for example. Similarly, system pressure limits must not be exceeded to avoid damage or increased maintenance costs.

In this paper, we explore the instantaneous and long-term performance of PTES in off-design conditions. We consider how ambient temperatures, storage tank temperatures, and load-following capabilities affect the power output and thermal conditions. In particular, we examine how storage tank temperatures and state-of-charge are affected and the impact on subsequent cycles. We develop an operational strategy that enables robust performance over a range of off-design conditions and will improve the reliability and value of the system.

1.1. Recent developments and literature review

Recent advances in PTES development include three notable small-scale test facilities. Firstly, researchers in the UK built and tested a concept [9] originally developed by Isentropic Ltd. [10] which uses reciprocating compressors/expanders and layered packed bed thermal storage [11]. The 150 kW$_e$ demonstrator was operated at part load and was estimated to achieve a round-trip efficiency of 57 % [9]. Secondly, Echogen installed a 100 kW$_{th}$ supercritical-CO_2 thermal cycle using sand and ice as the hot and cold storage materials, respectively, with the objective of demonstrating operation and control of the storage temperatures and CO_2 flow [12]. Full thermal cycles were not deployed: throttle valves were used instead of expanders/turbines and additional heating elements produced the required temperatures, but never-the-less, operation and control was tested, and stable component and system performance over multiple cycles was demonstrated [13]. This work has resulted in funding to demonstrate a 50 MW$_e$ sCO_2-PTES system with 12 h or storage in Healy, Alaska [14]. Thirdly, a 5 kW$_e$ test facility using an air-based cycle using thermal oil hot storage and water-glycol cold storage was deployed by South-West Research Institute [15]. This facility was used to investigate transient behavior and control methods during charge and discharge start-up. Several control strategies were used, including inventory control, turbomachinery speed control, and recycling working fluid and storage fluids, and the project demonstrated methods to manage cycle start-up and state-of-charge (SOC).

'Inventory control' is one method to achieve part-load operation of PTES [16]. Inventory control was developed for closed-cycle gas turbines [17] and has been recently applied to supercritical CO_2 cycles [18]. It operates by adding or removing working fluid from the system via a buffer tank. The objective of inventory control is to maintain a constant volumetric flow rate through the turbomachinery which is achieved by adjusting the mass flow rate and pressure – e.g. part-load power output is reached by reducing the mass flow and pressure linearly, thereby keeping the volumetric flow constant.

Previous work indicates that for gas-based cycles, inventory control keeps pressure ratios (and thus temperature ratios) close to their design values. For a system with liquid storage, temperatures at each point in the cycle remain constant throughout charge and discharge. As a result, the off-design round-trip efficiency and power output is similar to the design value over a wide range of part-load conditions [16]. Similar results were found for PTES using packed bed TES [19]: such storage systems complicate control because thermal gradients in the packed beds require careful management [11] and can impose variable temperatures on the turbomachinery. Ref. [19] demonstrated that inventory control could be used to maintain constant load despite significant temperature variations due to the packed beds. However, results did indicate that packed bed temperature changes do limit the controllability and efficiency of the discharging cycle. These affects can be reduced by limiting the exit temperature variation from the packed beds which requires careful design and management [20]. Transient models of packed-bed PTES were described in Ref. [21] described how packed bed exit temperatures change continuously and how hot and cold packed beds accrue and discharge working fluid at different rates, leading to different mass flow rates in different parts of the cycle, further complicating operation and emphasizing the importance of transient analysis for these types of PTES systems.

Albay et al. described an off-design model for a recuperated Brayton PTES system with liquid storage using CO_2 for the thermal cycle working fluid [22]. While inventory control was used to control power output, a key development was an algorithm to manage the storage tank SOC with the goal of maintaining equal SOC of the hot and cold tanks. (Unequal SOC on the hot and cold storage is referred to as being 'unbalanced' and is permissible. However, unbalanced tanks reduce the system energy capacity and re-balancing the tanks leads to system inefficiencies). The authors described a co-generation system where the hot storage could deliver heat as well as electricity. When delivering heat, the cold storage is not discharged, thereby leading to unbalanced storage tanks. The tank SOCs were re-balanced by introducing a balancing heat exchanger that effectively discharged the cold storage in proportion to the hot storage discharged for heating purposes.

Benato et al. developed an off-design model of electrically-charged packed bed TES which is later discharged through a gas turbine [23]. The system reaches higher temperatures (1200K) than those typically considered for PTES using a heat pump. This system also has the advantage of only using hot storage – so balancing the SOC of both hot and cold storage is not necessary. However, inventory control is not a viable operation strategy due to employing an open-gas cycle. The authors used compressor inlet guide vane control while maintaining constant rotational speed. They found that as packed bed discharged, the exit temperature decreased leading to a change in the system pressure ratio and a reduction in power output.

Dynamic models have also been developed by several research groups and are used to investigate transient response. For example, a dynamic model of a Joule-Brayton cycle with liquid storage was developed in Ref. [24]. Two scenarios were considered; warm-start and a sudden increase in power requested by the grid. These events were managed with inventory control, and the authors found that the system could respond and reach steady-state operation in less than 1 min, although real-world considerations may increase this time. Start-up was investigated in Ref. [25], and this requires a control strategy for the turbomachinery rotational speed. It was found that the time taken to achieve steady-state operation in charge was independent of the rate of increase of rotational speed. However, during discharge, the rate of increase of rotational speed affected both the start-up time and the stability of the rotational speed. The impact of step changes to the power input and output were considered in Refs. [26, 27], and the dynamic response was managed with inventory control. During charge, it was found that cycle temperatures remained constant when the power input reduced by 50 %, which made it possible to charge the storage tanks to the correct temperature [26]. Similarly, during discharge, a 50 % reduction in required load could be managed, and load following capability was demonstrated [27].

Pettinari et al. described a dynamic model of a recuperated Brayton heat pump delivering thermal energy at above 250 °C for industrial process heat applications [28]. The model accounted for controls, inertia, and volume dynamics of the various components and used two controllers: an anti-surge controller to prevent compressor surge and a temperature controller to modify the temperature delivered by the heat pump. Temperature control was achieved by varying compressor speed. In Ref. [28], the authors described cold start-up and quantified start-up time to be around 2 h, although this could be reduced by increasing the allowable rate of temperature change in the heat exchangers. The researchers also examined whether this heat pump could deliver different sink temperatures in response to load requirements [29]. Although the outlet temperature oscillates slightly, it settled on the required temperature within timeframes that were acceptable to industrial needs. Typically, larger temperature changes took longer to settle – e.g. a 5 °C increase could be achieved within 8.5 min but a 20 °C increase took 40 min. The heat pump was designed for industrial process heat applications, and therefore did not later convert heat to electricity in a heat engine. Similar techniques could be used in a Carnot Battery where the generated heat is later converted into electricity. However, varying compressor speed to change the hot storage temperature could affect the cold storage temperature or SOC, and the off-design impacts on the discharge cycle have to be understood.

An emerging class of PTES systems are Thermally-Integrated PTES (TI-PTES) which integrate other thermal sources/sinks. For instance, waste heat or low-temperature solar thermal can be used as the heat pump source, thereby improving the Coefficient of Performance. Off-design models of a TI-PTES with solar thermal are described in Ref. [30]. Such systems do not have cold storage and so avoid challenges associated with balancing the storage tanks. However, a detailed control system is required to manage the variable mass flow and temperature delivered by the solar field. The authors considered several days of operation and described how the system responded to variable input conditions. An alternative TI-PTES design was presented in Ref. [31] and dynamic models developed. The models illustrated the importance transient variations in solar energy and temperature by considering four representative days in China.

1.2. Research gaps addressed in this paper

The existing literature has mainly focused on either part-load operation or dynamic response to changes in environmental conditions or during start-up or. For closed gas cycles, part-load operation is facilitated by inventory control, and previous work has demonstrated the capability of inventory control to manage variations in power input and output, maintain high efficiencies in these part-load events, and keep the storage tanks close to their design temperatures. However, other external factors may affect tank temperatures – such as heat leakage and ambient temperatures. Therefore, in this paper we explore the impact of ambient temperature, storage tank temperatures, and load-following capabilities in off-design scenarios. We demonstrate that inventory control can be used to manage these variations while delivering the design power rating even at off-design conditions and note this may lead to pressures that exceed the rated design point.

An important consideration is how off-design operation impacts storage tank temperatures and state-of-charge (SOC) – and how this may change over several cycles. Changing storage tank temperature may reduce power output or efficiency. Unequal SOC of the hot and cold tank leads to reduced energy capacities and requires an energy penalty to rectify. Any change to tank temperature or SOC will also influence subsequent cycles. Previous work in the literature has not fully explored these aspects of PTES operation. In this paper, we develop an operational strategy that enables robust performance over a range of off-design conditions, and in particular, consider how tank temperature and SOC are impacted. We introduce control of the storage fluid mass flow rates, which enables flexible management of the tank temperature and SOC. This is important for robustly managing the long-term impacts of off-design operation.

A Joule-Brayton PTES system with liquid storage is introduced in Section 2. The methods for modelling the design and off-design behavior of components are then described in Section 3. Strategies for controlling storage tank temperature and state-of-charge and for maintaining stable operating points are discussed, and an off-design control strategy is outlined. The control strategy includes inventory control and also management of storage fluid mass flow rates. This strategy is then applied to several variables in Section 4 to demonstrate the system performance and response to off-design conditions. Parameters of particular interest are the ambient temperature and storage tank temperatures. It is also demonstrated that the working fluid mass flow rate is a controllable off-design parameter that may be used to achieve a desired power input or output. Numerous consecutive charge-discharge cycles are evaluated in some cases to illustrate the longer-term implications of some off-design parameters.

2. System design

PTES systems integrate thermodynamic cycles with thermal energy storage (TES). The charge phase consists of a heat pump that uses electricity to move energy from cold thermal energy storage to hot thermal energy storage. Electricity is later generated in the discharge phase by a heat engine operating between the hot and cold thermal reservoirs.

A Joule-Brayton PTES with liquid TES is illustrated in Fig. 1 and corresponding temperature-entropy diagrams are shown in Fig. 2. The thermal cycle working fluid (nitrogen) is kept in the gas phase and is compressed and expanded using axial turbomachinery. Heat is transferred between the thermal cycles and storage fluids using counter-flow heat exchangers. The hot and cold storage each comprise two tanks: a ‘source’ tank and a ‘sink’ tank. During charge, fluid is removed from the source tank, passes through a heat exchanger, and is delivered to the sink tank. During discharge, the fluid is removed from the sink tank and delivered to the source tank.

The charging phase is a heat pump and operates as follows: the working fluid is first compressed ($1 \rightarrow 2$) to high temperature and pressure and then enters the hot heat exchanger where heat is transferred ($2 \rightarrow 3$) to the hot storage fluid. Correspondingly, the hot storage fluid travels from the source tank to the sink tank and is heated from $T_{\mathrm{source,h}}$ to $T_{\mathrm{sink,h}}$. The working fluid enters the high-pressure side of a recuperator ($3 \rightarrow 4$) and is then expanded to a low temperature in a turbine ($4 \rightarrow 5$). Heat is extracted from the cold storage fluid which passes from the cold source tank to cold sink tank and is cooled from $T_{\mathrm{source,c}}$ to $T_{\mathrm{sink,c}}$, and in the process warms the working fluid ($5 \rightarrow 6$). The fluid is returned to the original pressure and temperature on the low-pressure side of the recuperator.

The discharging phase is a heat engine and operates by reversing the direction of each fluid. The same heat exchangers are used but a different compressor and turbine is employed. The working fluid is compressed ($1' \rightarrow 2'$) and heated in the high-pressure side of the recuperator ($2' \rightarrow 3'$) and hot heat exchanger ($3' \rightarrow 4'$). The hot storage fluid therefore flows back from the sink tank to the source tank. The working fluid is then expanded to generate power ($4' \rightarrow 5'$) before being cooled in the recuperator ($5' \rightarrow 6'$), heat rejection air cooler ($6' \rightarrow 7'$), and cold heat exchanger ($7' \rightarrow 1'$). The cold storage fluid returns to the source tank from the sink tank.

Storage fluids must remain liquid over the temperature range of operation – and should not freeze, become excessively viscous, evaporate, decompose, or combust. Nitrate molten salt (i.e. “Solar Salt” – a blend of sodium nitrate and potassium nitrate) is chosen for the hot storage. Solar Salt can be stored and operated up to 565 °C as demonstrated in the Concentrating Solar Power (CSP) industry [32]. The molten salt should be kept above 250 °C to prevent freezing and the thermal cycles therefore include a recuperator which enables the system to operate over a wide temperature differential while managing the more limited operating temperature range of the molten salts. Alternative hot storage fluids include thermal oils and other molten salt blends. Thermal oils could avoid the need for the recuperator but have lower maximum temperatures that reduce the round-trip efficiency and increase the cost and Levelized Cost of Storage (LCOS) [33]. On the other hand, some molten salt blends (such as chloride molten salts) can operate at higher temperatures (up to 750 °C) and while this increases the efficiency, the hot heat exchanger and recuperator must then use nickel alloys which leads to higher costs. Analysis in Ref. [33] suggested that there was a small LCOS improvement for PTES with chloride molten salts, but these improvements were small compared to the uncertainty.

Results for a nominal design point are presented in Table 1 and Fig. 2. This design is based on an ‘optimal’ design identified in Ref. [33] and represents a balance between maximizing the round-trip efficiency and minimizing the LCOS. Results differ slightly from those in Ref. [33] due to a small difference in the design. Heat is rejected from the system in an air-cooler, and the PTES configurations in Ref. [33] rejected heat in both the charge and discharge phases. Here, the design is simplified by only rejecting heat in the discharge phase.

3. Methods

3.1. Design methods

Design-point PTES calculations use methods developed in Ref. [33] and the same computational code is used for this article. The methods

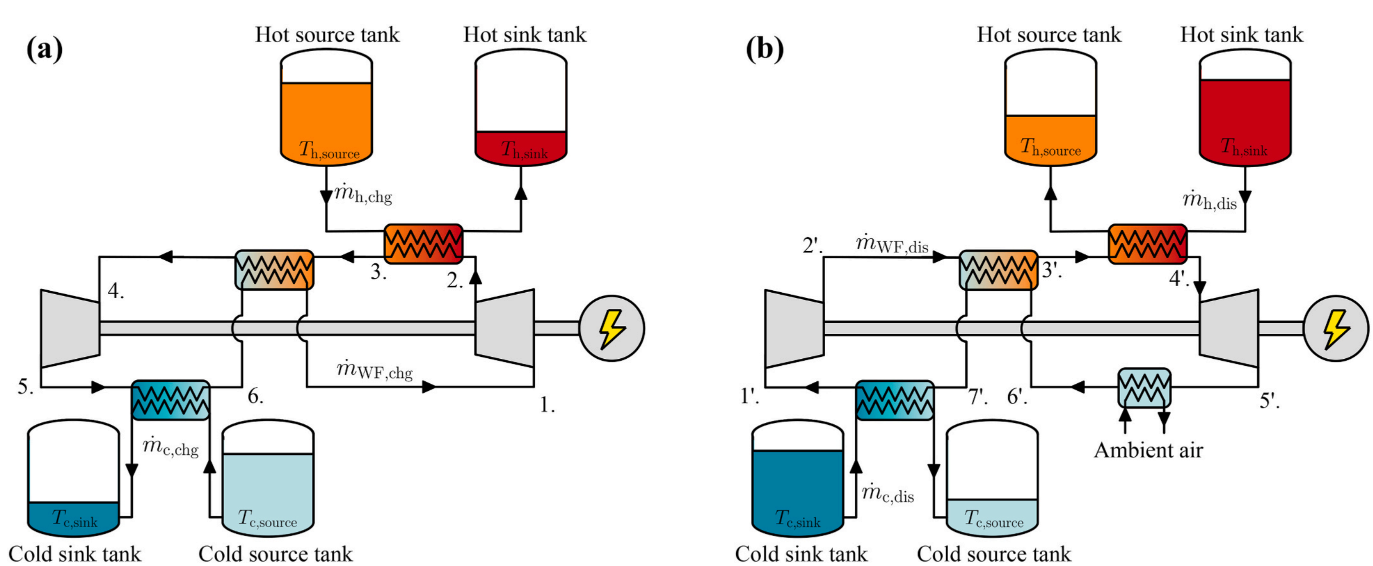

Fig. 1. Schematic of liquid-storage Joule-Brayton PTES (a) charge phase (b) discharge phase. The same storage tanks and heat exchangers are used in charge and discharge but different turbomachinery trains are used.

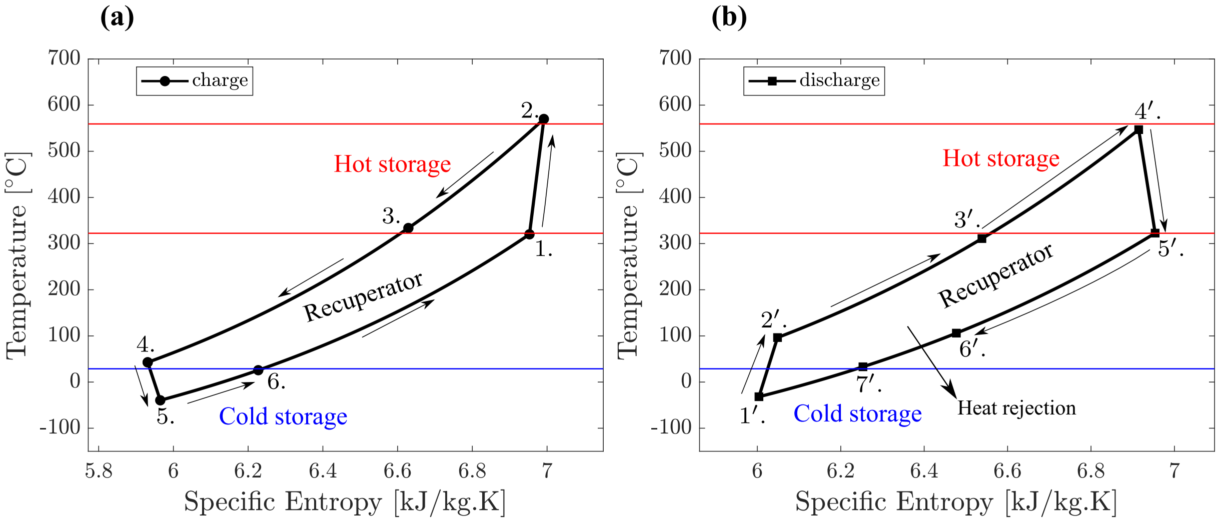

Fig. 2. Temperature-entropy diagrams of PTES (a) charge (b) discharge. Numbers correspond to Fig. 1.

Table 1
Main design parameters and results for a PTES system based on optimal result 'N2' from Ref. [33].

		Charge phase	Discharge phase
T_{amb}	°C	25.0	25.0
T_1 or $T_{5'}$	°C	325.0	325.7
p_1 or $p_{5'}$	bar	7.9	8.0
Pressure ratio	–	3.1	3.9
$\dot{m}_{WF}$	kg/s	823.2	823.2
$\dot{m}_h$	kg/s	600.3	606.0
$\dot{m}_c$	kg/s	365.8	369.3
$T_{source,h}$	°C	326.0	326.0
$T_{sink,h}$	°C	565.4	565.4
$T_{source,c}$	°C	28.9	28.9
$T_{sink,c}$	°C	−40.9	−40.9
$\dot{W}$	MW_e	−160.5	100.0
τ	h	10.2	10.0
COP	–	1.36	–
η_{HE}	–	–	45.4
η_{RT}	%	61.7	

are briefly recapped here, and an overview of the calculation procedure is shown in Fig. 3. Components are typically characterized by a performance metric, such as efficiency or pressure loss, and the nominal performance metrics used in this work are summarized in Table 2.

Separate sets of turbomachinery are used for the heat pump and heat engine; the system requires two compressors and two expanders. It is assumed that the compressor and turbine are on the same shaft and therefore rotate at the same speed. Turbomachinery design performance is characterized by a polytropic efficiency η_p. For perfect gases, the compressor and turbine temperature ratio and pressure ratio are related by:

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma\eta_p}} \tag{1}$$

$$\frac{T_4}{T_5} = \left(\frac{p_4}{p_5}\right)^{\frac{\eta_p(\gamma-1)}{\gamma}} \tag{2}$$

Where the numbering corresponds to the charging system shown in Fig. 2a, and $\gamma = c_p/c_v$. For the PTES systems considered here, the variation in temperature and pressure are large enough to appreciably change the value of γ. Therefore, Eqs. (1) and (2) are integrated numerically and working gas properties are calculated using CoolProp [34]. The heat pump shaft is driven by a motor while the heat engine drives a generator. Electrical and mechanical losses occur in these devices which are represented by an efficiency term $\eta_{motor/gen}$ [35].

Heat exchanger performance is characterized by an effectiveness $\varepsilon = \dot{Q}/\dot{Q}_{max}$ and fractional pressure loss $f_p = \Delta p/p_{in}$. The hot and cold heat exchanger and the recuperator are modelled as counterflow heat exchangers with circular channels, and all have the same value of ε_{HX} and $f_{p,HX}$. The air cooler is modelled as a cross-flow heat exchanger and has an individual value of ε_{cooler} and $f_{p,cooler}$. For the charging design point, heat exchangers have balanced flow – i.e. the hot and cold streams have equal values of $\dot{m}c_p$. Storage fluid properties are interpolated from a data table.

The same heat exchangers are used in the charge and discharge phase, and small differences in cycle pressure and temperature mean that the heat exchangers operate in off-design mode during discharge. To adequately evaluate heat exchanger off-design performance the geometry must be calculated. A first estimate of the charging cycle state points and mass flow rates is calculated using target values of ε and f_p, see Fig. 3. The resulting conditions are used as inputs to a heat exchanger model which calculates frictional pressure losses and heat transfer coefficients at discretized points along the length of the heat exchanger. For turbulent flow, appropriate equations for the friction coefficient c_f and Nusselt number Nu are [36]

$$c_f = (1.58 ln(Re))^{-2} \tag{3}$$

$$\mathrm{Nu} = \frac{0.5 c_f Pr(Re - 1000)}{1 + 12.7(Pr^{2/3} - 1)(c_f/2)^{1/2}} \tag{4}$$

Where Re is the Reynolds number and Pr is the Prandtl number.

An iterative procedure calculates the heat exchanger length and heat transfer area necessary to achieve the target values of ε and f_p, and this process is described in detail in Ref. [33]. As shown in Fig. 3, the charge cycle state points are then re-calculated using the discretized heat exchanger model. This process is iteratively repeated until the state points converge on consistent values.

The discharge cycle also iteratively calculates the cycle state points. The heat exchanger geometries calculated in the charging phase are used as inputs to the off-design heat exchanger model, which evaluates c_f and Nu along the discretized heat exchanger length using Eqs. (3) and (4). An iterative process ensures the heat transfer balance is satisfied, as detailed in Ref. [33]. Rather than assuming the heat exchangers are balanced, the hot storage fluid mass flow rate $\dot{m}_h$ is varied to ensure that the fluid is returned to the temperature it had at the start of charge. The

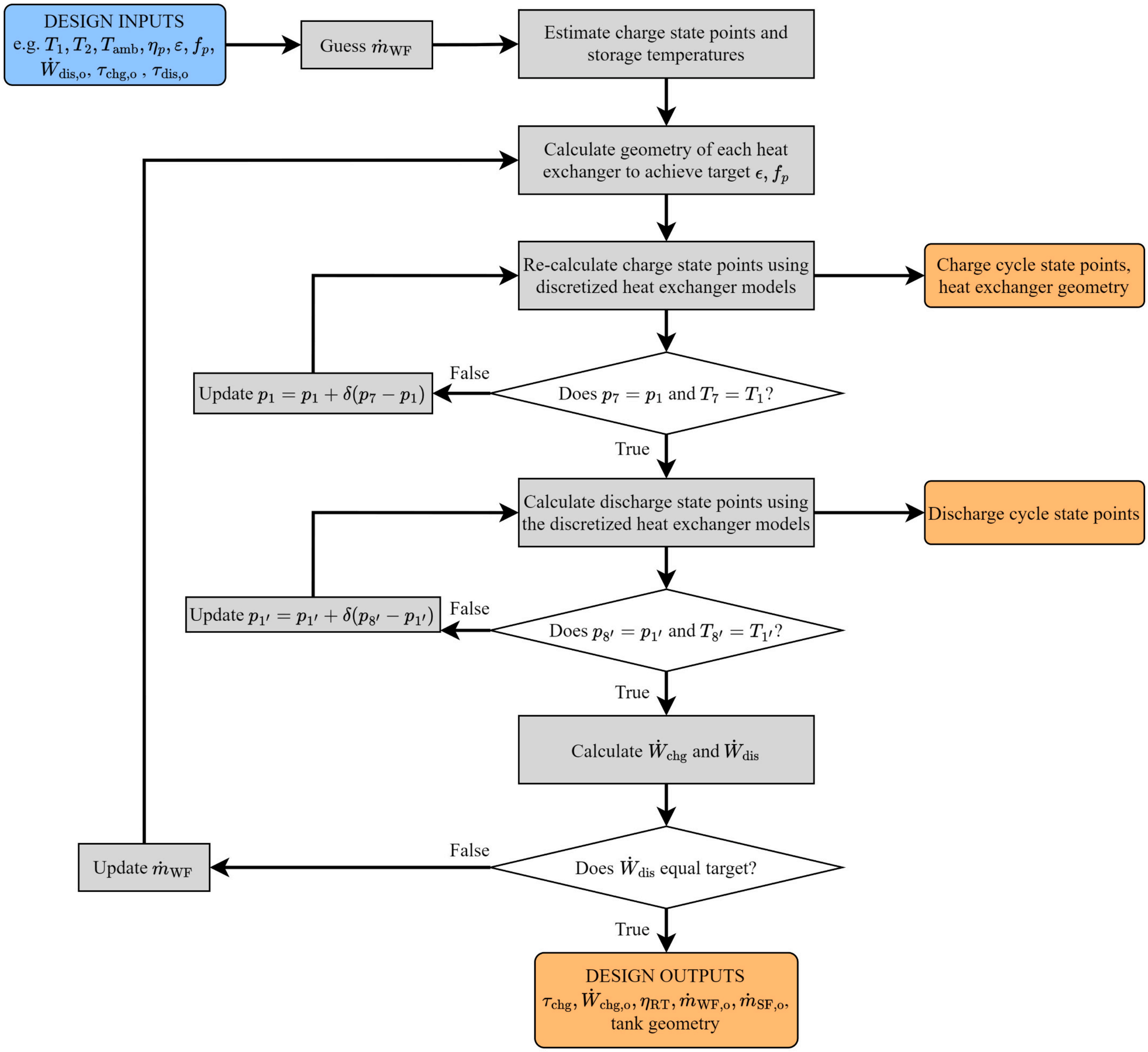

Fig. 3. Flow diagram illustrating the calculation process for design-point PTES performance evaluation.

Table 2
Nominal performance parameters for PTES based on optimal design 'N2' from Ref. [33].

Parameter		Value
η_p	%	90.0
ε_{HX}	%	98.0
$f_{p,HX}$	%	0.7
$\varepsilon_{\mathrm{cooler}}$	%	90.0
$f_{p,\mathrm{cooler}}$	%	0.5
η_{pump}	%	80.0
η_{fan}	%	75.0
$\eta_{\mathrm{motor/gen}}$	%	98.0
Working fluid		Nitrogen
Hot storage fluid		Nitrate molten salt ($NaNO_3$-KNO_3)
Cold storage fluid		Methanol

cold storage fluid mass flow rate is varied in proportion to $\dot{m}_h$ to ensure the hot and cold tanks discharge at the same rate.

This heat exchanger model was developed in Ref. [33] and was validated against results from a CFD simulation for a range of off-design operating conditions (as presented in the Appendix of that work). The heat exchanger model showed good agreement which supports its usage in the PTES off-design model that is described in Section 3.2.

Once the cycle state points are calculated for both charge and discharge, the net work input and output can be evaluated. The net work includes the thermodynamic work of the compressors and expanders as well as losses due to the motor, generator, and pumps. Pumps are required to move storage fluids, while a fan blows air through the air cooler. These are characterized by an efficiency term $\eta_{\mathrm{pump/fan}}$ which includes the efficiency of the pump's motor. The pump and fan power inputs are given by the power required to overcome the fluid pressure losses and motor inefficiency, such that the work input is given by, for example,

$$\dot{W}_{\text{pump}} = \frac{\dot{m}\Delta p}{\rho \eta_{\text{pump}}} \tag{5}$$

The charging work input $\dot{W}_{\text{chg}}$ and discharging work output $\dot{W}_{\text{dis}}$ are given by

$$\dot{W}_{\text{chg}} = \left(\dot{W}_{\text{comp}} - \dot{W}_{\text{exp}}\right) \Big/ \eta_{\text{motor}} + \dot{W}_{\text{pump}} \tag{6}$$

$$\dot{W}_{\text{dis}} = \left(\dot{W}_{\text{exp}} - \dot{W}_{\text{comp}}\right) \eta_{\text{gen}} - \dot{W}_{\text{pump}} + \dot{W}_{\text{fan}} \tag{7}$$

The model specifies a target net work output (of 100 MW_e in this work). Another level of iteration is required to find the working fluid mass flow rate $\dot{m}_{\text{WF}}$ that delivers the required power output. (Note, the charge and discharge cycles have the same value of $\dot{m}_{\text{WF}}$ which leads to approximately equal charge and discharge durations). Having established the mass flow rate of the working fluid and storage fluids (via the heat exchanger calculations), the storage system is sized.

Storage tanks are modelled using a double-tank arrangement; there is a source tank and sink tank for both hot and cold storage. Insulation thickness is calculated assuming a 0.5 % energy loss per day, but these energy losses are not considered when calculating the round-trip efficiency since the total energy losses will depend on the exact charge-discharge schedule. Mixing can occur in the tanks (e.g. if fluid of a different temperature is introduced into the tank). In this work, mixing losses are minimized by ensuring the tanks fully empty and fill during each phase.

The discharge duration τ_{dis}, is specified (10 h in this work) and the mass of hot and cold storage fluid are then given by $\dot{m}_h \tau_{\text{dis}}$ and $\dot{m}_c \tau_{\text{dis}}$, respectively. The charging duration τ_{chg} is then calculated by finding the time taken move this mass of fluid at the charging mass flow rate.

The round-trip efficiency η_{RT} is calculated by considering the fraction of input electrical energy that is recovered during discharge, and is expressed as:

$$\eta_{RT} = \frac{\dot{W}_{\text{dis}}\, \tau_{\text{dis}}}{\dot{W}_{\text{chg}}\, \tau_{\text{chg}}} \tag{8}$$

Typically, this equation is evaluated after a single charge-discharge cycle. When considering off-design operation, it can be helpful to track how η_{RT} changes over time and also to consider the average value over multiple cycles.

The performance of the charging and discharging phases may also be individually assessed using either the work input/output, or by using the heat pump Coefficient of Performance (COP) or the heat engine efficiency, η_{HE} – e.g.:

$$\text{COP} = \frac{\dot{Q}_{h,\text{chg}}}{\dot{W}_{\text{chg}}} \tag{9}$$

$$\eta_{HE} = \frac{\dot{W}_{\text{dis}}}{\dot{Q}_{h,\text{dis}}} \tag{10}$$

Where $\dot{Q}_h$ is the heat added or removed from the hot storage – e.g. $\dot{Q}_{h,\text{chg}} = \dot{m}_{WF}(h_2 - h_3)$ during charge. Since heat losses are negligible, it should be the case that $\dot{Q}_{h,\text{dis}} = \dot{Q}_{h,\text{chg}}$ as long as the tank is fully discharged. However, scenarios can arise where the tanks are not fully discharged, in which case the energy balance over multiple charge-discharge cycles should be considered to properly evaluate the performance.

3.2. Off-design models

Off-design models of each component have also been developed and are described in detail in Ref. [33].

Turbomachinery behavior follows characteristic curves which relate the inlet conditions to the pressure ratio and efficiency. The inlet conditions can be characterized using the 'swallowing capacity' or mass flow coefficient $\dot{m}_c$:

$$\dot{m}_c = \frac{\dot{m}\sqrt{T}}{p} = \frac{\dot{m}_o\sqrt{T_o}}{p_o} = \text{constant} \tag{11}$$

Example curves are shown in Fig. 4 using data from Refs. [37, 38] in terms of the normalized mass flow coefficient $\dot{m}_r = \dot{m}_c/\dot{m}_{c,o}$. The shape of these curves is consistent with those shown in other works, and the shape can be adjusted to model specific machines by modifying the coefficients. It should be noted that the operation strategy employed here aims to keep the rotational speed and mass flow coefficient constant. Therefore, the chosen characteristic curves have a small impact on the results, since $\dot{m}_r$ is kept close to 1, and the pressure ratio and efficiency does not deviate significantly from the design values, as shown in Section 4.

The motor/generator also has a characteristic curve, which defines efficiency as a function of load. A curve was obtained from Ref. [35] and this curve indicates that the efficiency only drops significantly below its design value for loads below 50 % of the design value.

Heat exchanger models were developed in Ref. [33]. As described in Section 3.1, the heat exchanger geometry is calculated to achieve a design effectiveness and pressure loss. The heat exchanger is discretized, and flow properties, heat transfer, and pressure losses are calculated for each section, given the off-design inlet properties. An iterative procedure is implemented to calculate the outlet flow properties and ensure the total heat transfer is achievable with the available geometry. The user has flexible control over the heat exchanger behavior. For example, if the mass flow rates of both hot and cold stream are specified then the resulting outlet properties are calculated. Alternatively, the user can specify a target outlet temperature for one of the streams, and the model then calculates the mass flow rate necessary to achieve this.

The air cooler fan power is calculated based on the air mass flow rate which varies depending on the ambient temperature. For ambient temperatures below the design point, the air mass flow rate is reduced to maintain a constant temperature into the cold heat exchanger $T_{7'}$. For ambient temperatures greater than the design value, the air mass flow rate varies in proportion to the working fluid mass flow rate – in this study, $\dot{m}_{air} = 2\dot{m}_{WF}$.

3.3. Off-design control

PTES components may operate in off-design conditions due to factors beyond the operator's control (such as ambient temperature variations, or heat leakage/addition to the storage tanks), or due to decisions made by the operator (such as varying the mass flow rates of the working fluid or storage fluids). These off-design impulses impact two sub-systems: (1) The thermodynamic cycle (heat pump or heat engine); (2) The storage tanks. Impacts to the thermodynamic cycle are instantaneous whereas changes to storage tank temperatures affect the next charging or discharging phase.

The PTES off-design strategy implemented in this article comprises two control systems: (1) Inventory control to manage turbomachinery constraints, maintain cycle temperatures, and load follow (2) Storage fluid control to manage the temperature and state-of-charge (SOC) of the storage fluid. The temperature of the storage fluid affects the thermodynamic cycle state points, and vice versa, meaning that there are interactions between the two control systems and iteration is required to find a converged operating point for each set of off-design conditions.

3.3.1. Inventory control

Inventory control is predominantly a tool for providing load-following (or part-load) capability. However, it can also be used to maintain stable cyclic behavior when off-design conditions affect cycle temperatures.

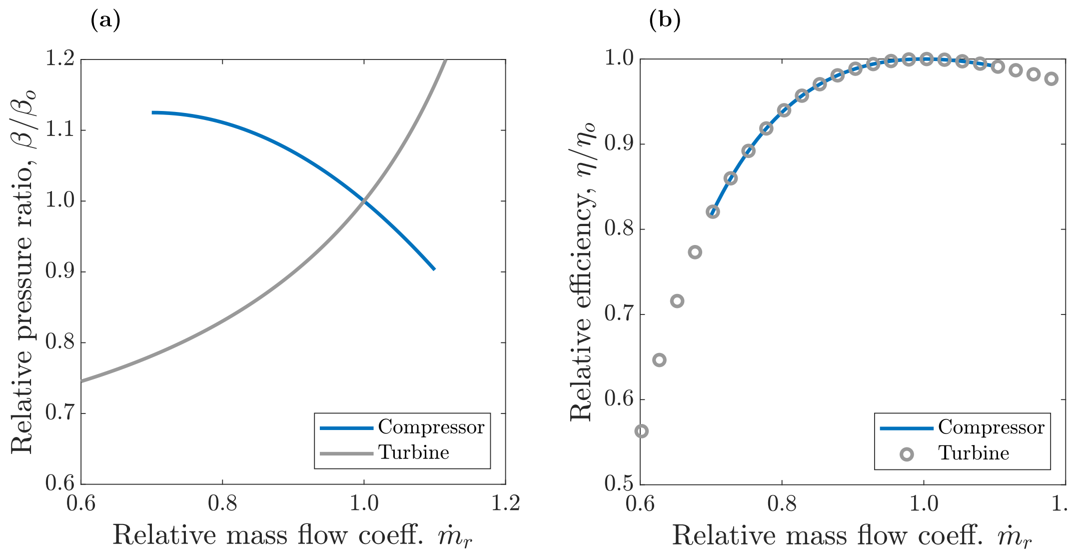

Fig. 4. Off-design characteristic curves for compressors and turbines. Data from Refs. [37, 38].

Inventory control varies the mass of working fluid within the cycle in order to adjust the system pressure. Fig. 4 illustrates characteristic curves of turbomachinery efficiency and pressure ratio as a function of the normalized mass flow coefficient $\dot{m}_r$ (Eq. (11)), which is itself a function of the turbomachinery inlet temperature, pressure, and mass flow rate. Adjusting these properties to keep $\dot{m}_r$ close to its design value enables the cycle to be operated close to its design pressure ratio, temperatures, and efficiency. For example, part-load is achieved by reducing the mass flow rate, and Eq. (11) indicates that reducing the system pressure proportionally will result in $\dot{m}_r = 1$: essentially, this maintains a constant volumetric flow rate through the turbomachine, as reducing pressure corresponds to reducing the density, which balances the reduced mass flow rate. Similarly, if the turbomachine inlet temperature changes (as a result of storage tank temperature variation, for example), then the pressure can again be varied to achieve $\dot{m}_r = 1$.

A further consideration is that these PTES systems use closed thermodynamic cycles, which require that the system returns to its original thermodynamic state – i.e. there is no change in state functions such as internal energy or entropy over the course of the cycle. For a simplified system with no pressure losses, this means that the compressor and turbine must have equal pressure ratios. When considering off-design operation, these pressure ratios must continue to equal one another so that cycle state-points remain constant and do not change over time. Fig. 4 shows that the compressor and turbine pressure ratio have opposite dependence on $\dot{m}_r$ – increasing $\dot{m}_r$ leads to larger pressure ratios in the turbine and reduced pressure ratios in the compressor. Thus, varying system pressure to maintain $\dot{m}_r = 1$ is an important feature that enables controllable off-design operation of these closed thermodynamic cycles. In reality, the pressure ratios are not exactly equal due to pressure losses in the heat exchangers. This is complicated further during off-design operation, as the heat exchanger pressure losses vary. Further complications arise if off-design storage temperatures may also affect one turbomachine and not the other. Therefore, an iterative procedure that modifies both the compressor inlet temperature and pressure is implemented. This procedure primarily ensures that pressure drops balance around the cycle and the cycle is returned to its original thermodynamic state.

3.3.2. Storage tank control

The mass flow rate of the hot and cold storage fluids ($\dot{m}_h$ and $\dot{m}_c$, respectively) can be varied independently. This control has two consequences: Firstly, it affects the heat exchanger exit temperature and can therefore be used to heat or cool the storage fluids to the correct temperatures in each phase. This ensures that the thermodynamic cycle in the subsequent phase will be operated at the correct temperatures and keeps the storage fluids within their operational temperature range. Secondly, it changes the time taken to fully charge or discharge the tanks, which can 'unbalance' the hot and cold storage – i.e. the hot tanks could be fully charged before the cold tanks, or vice versa (and similar effects could occur in discharge). Design operation will not be possible in the next phase because one set of tanks contains less energy than required. An example to fully illustrate this is provided in Appendix A.

While a variety of storage tank control methods are feasible, the strategy chosen here is to vary $\dot{m}_h$ to maintain the design temperature of the hot fluid in charge and discharge. To keep the tanks balanced, $\dot{m}_c$ is varied in proportion to $\dot{m}_h$ which means that the temperature of the cold fluid delivered to the tank deviates from the design value. This creates off-design conditions in future phases, but results show that these effects are relatively small and self-correcting, as demonstrated in Section 4, and are less significant than allowing the tanks to become unbalanced.

3.3.3. PTES off-design control strategy

The off-design control strategy combines inventory control with storage tank control and aims to find operating conditions that lead to a stable, closed thermodynamic cycle and provide robust control of storage tank temperatures and SOCs:

1. Vary the system pressure using inventory control to ensure the turbomachinery is operated close to its design $\dot{m}_r$ and that the cycle closes.
2. During the charge phase:
   a. Vary $\dot{m}_h$ to ensure the hot sink temperature is kept as close to the design value as possible.
   b. Vary $\dot{m}_c$ such that $\dot{m}_c = \dot{m}_{c,o} \cdot \dot{m}_h / \dot{m}_{h,o}$
3. During the discharge phase:
   a. Vary $\dot{m}_h$ to ensure the hot source temperature is kept as close to the design value as possible.
   b. Vary $\dot{m}_c$ such that $\dot{m}_c = \dot{m}_{c,o} \cdot \dot{m}_h / \dot{m}_{h,o}$

Constraints 2a. and 3a. avoid decomposition or freezing of the molten salt, respectively, while constraints 2b. and 3b. ensure that the cold tank SOC is equal to the hot tank SOC. This operation can lead to cold fluid temperatures that deviate from the design point, but these variations are found to be small and within the operating temperature envelope. For some off-design conditions, it may not be possible to keep the hot fluid at the design temperature due to other constraints in the thermal cycle, so the controller aims to minimize the difference between the off-design and design temperature.

The off-design controller uses an iterative approach to find a converged solution because inventory control can affect storage tank temperatures, and storage tank control can affect thermal cycle state points. The iterative approach for the charging phase is illustrated in Fig. 5 and a similar procedure is followed for the discharging phase. The design point model is run first as described in Fig. 3 and Ref. [33], and this specifies the geometry of each heat exchanger which is then used in the off-design model. The off-design model can then be executed once the off-design input variables (ambient temperature, tank temperatures, and working fluid mass flow rate) are specified. The off-design cycle state points are first estimated by assuming the temperatures are the same as the design values, and that the pressure at each point is scaled by the ratio of mass flow rates – i.e. $p_1 = p_{1,o}\, \dot{m}_{WF}/\dot{m}_{WF,o}$, as this maintains a constant mass flow coefficient. The off-design performance of each component is then evaluated until conditions at all cycle points have been estimated. For convergence, the thermal properties at State 7 should equal those at State 1 during charge. If not, the properties at State 1 are updated by evaluating a smoothed average of the properties at State 1 and 7: this ensures the model smoothly approaches a converged solution. In this work, the maximum system pressure is not constrained. The control strategies developed show the expected maximum pressure, and the designer can then determine an appropriate operating envelope

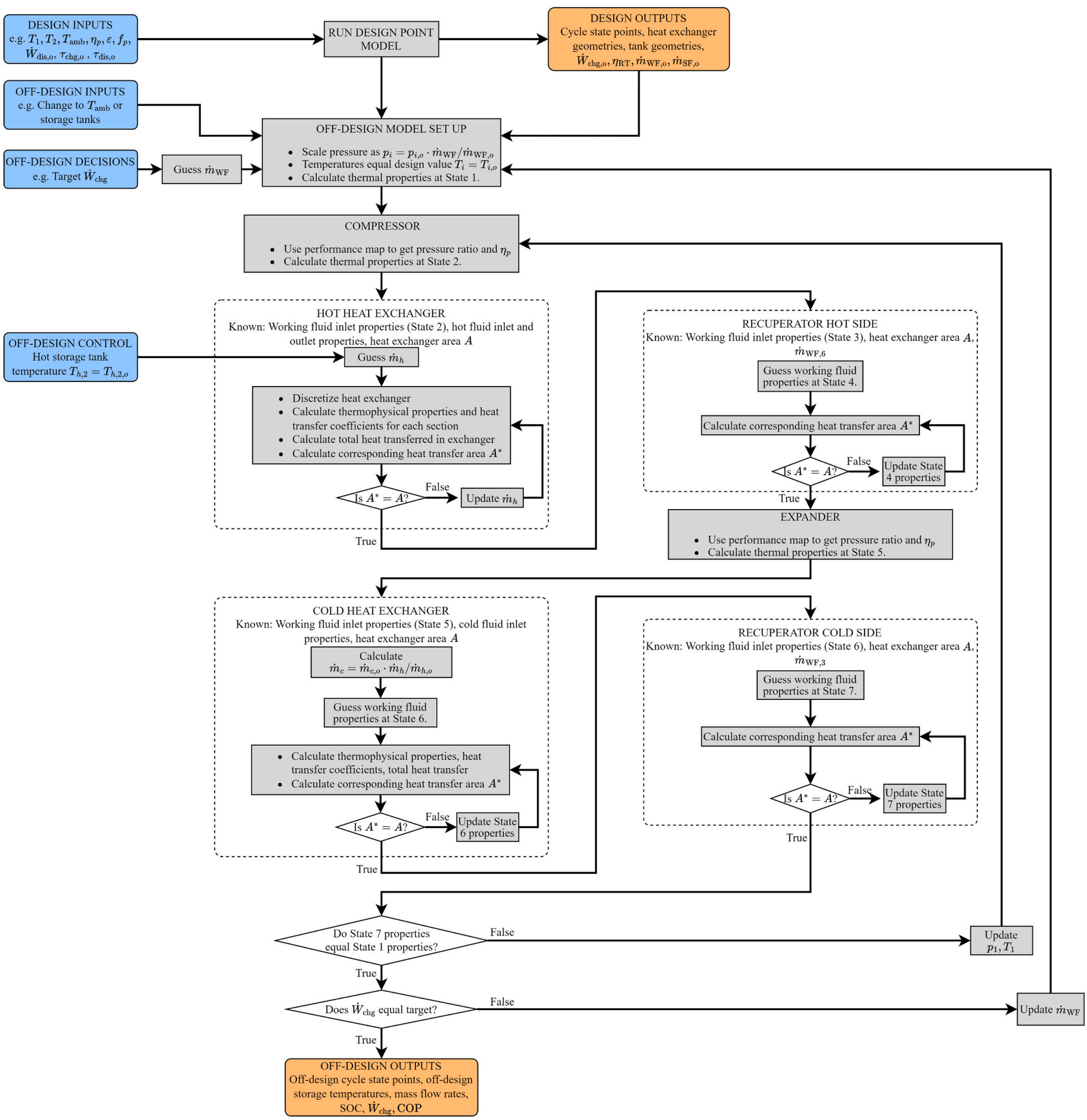

Fig. 5. Flowchart illustrating the process of identifying the off-design operating point for the charging phase.

based on the rated design.

It is assumed here that off-design conditions remain constant for the entirety of each charge or discharge phase. In reality, conditions (such as the ambient temperature) are likely to vary over the course of a phase. However, the aim of this work is to illustrate the broad impact of off-design conditions on the phase performance and future work will analyze more complicated, realistic load cycles. Each phase continues until one of the storage tanks is fully discharged, at which point the phase ends and its duration is calculated. After completion of a charge-discharge cycle, the full cycle performance may be evaluated in terms of the round-trip efficiency or other metrics. It is important to realize that the discharge phase may move the source tanks to off-design conditions and therefore impact the next charge phase, in which case several consecutive charge-discharge cycles should be evaluated to understand the full impacts of a single off-design impulse.

4. Results

In this section, several variables are moved to off-design values. The impact on the cycle performance is then quantified. Variables of interest include the mass flow rate during charge and discharge, the ambient temperature, and the temperature of the four storage tanks.

4.1. Heat pump mass flow rate

The plant operator has freedom to choose whether to vary the working fluid mass flow rate $\dot{m}_{WF}$ – this parameter is not affected by external factors. The power input or output has a linear relationship with $\dot{m}_{WF}$, therefore, PTES could flexibly deliver different charge/discharge rates depending on demand or electricity prices.

Fig. 6 shows the impact of $\dot{m}_{WF}$ on the power input and duration of the charging phase, and a strong linear relationship between $\dot{m}_{WF}$ and $\dot{W}_{\mathrm{chg}}$ is evident. The charge duration τ_{chg} is limited by the storage tank capacity which is sized based on the design $\dot{m}_{WF,o}$. Therefore, τ_{chg} varies inversely with $\dot{m}_{WF}$. Previous analysis of PTES dispatch in Californian electricity markets indicates that there are fewer opportunities to charge PTES than discharge – i.e. there are fewer low-price electricity hours than high-price electricity hours [39]. As a result, it is advantageous for PTES to be able to charge at a faster rate than discharge ($\tau_{\mathrm{chg}} < \tau_{\mathrm{dis}}$) so that it can optimize the discharging and charging price.

Fig. 6 demonstrates how a fixed PTES design could increase its charging rate by increasing $\dot{m}_{WF}$. Due to the inventory control strategy, the system pressures increase linearly with $\dot{m}_{WF}$ while pressure ratios remain roughly constant – see Table 3. Therefore, increasing the charging rate requires the turbomachinery, heat exchangers, pipes, valves, and instrumentation to be rated for pressures greater than the design value. For example, reducing τ_{chg} from 10 h to 6.67 h requires the maximum pressure to increase from 24.9 bar to 37.2 bar. This approach also requires the motor power to be increased by 50 %. While adding

Table 3
State points, tank temperatures, and turbomachinery performance for the charge phase with different working fluid mass flow rates.

	$\dot{m}/\dot{m}_o = 0.5$		$\dot{m}/\dot{m}_o = 1.0$		$\dot{m}/\dot{m}_o = 1.5$	
Variable	T, °C	p, bar	T, °C	p, bar	T, °C	p, bar
State 1.	322.8	4.0	325.0	7.9	326.0	11.8
State 2.	566.8	12.5	570.0	24.9	571.7	37.2
State 3.	333.1	12.4	331.1	24.8	331.9	37.1
State 4.	39.7	12.4	36.8	24.7	37.4	37.0
State 5.	−39.5	4.0	−42.4	8.0	−42.6	12.0
State 6.	27.4	4.0	27.4	8.0	26.8	11.9
Hot source tank	326.0		326.0		326.0	
Hot sink tank	565.2		565.3		565.6	
Cold source tank	28.9		28.9		28.9	
Cold sink tank	−39.0		−40.8		−40.4	
Comp. pressure ratio	3.1468		3.1461		3.1472	
Turbine pressure ratio	3.0829		3.0860		3.0933	
Comp. efficiency	89.9959		90.0000		89.9994	
Turbine Efficiency	89.9964		90.0000		89.9997	

Table 4
State points, tank temperatures, and turbomachinery performance for the discharge phase with different working fluid mass flow rates.

	$\dot{m}/\dot{m}_o = 0.5$		$\dot{m}/\dot{m}_o = 1.0$		$\dot{m}/\dot{m}_o = 1.5$	
Variable	T, °C	p, bar	T, °C	p, bar	T, °C	p, bar
State 1.	−36.0	3.9	−38.2	7.9	−36.3	11.7
State 2.	95.7	15.7	92.4	31.4	96.2	46.8
State 3.	317.7	15.7	318.9	31.4	316.9	46.8
State 4.	562.8	15.6	560.5	31.3	559.9	46.7
State 5.	326.9	4.0	325.7	8.0	324.3	11.9
State 6.	103.8	4.0	96.4	8.0	99.7	11.8
State 7.	32.0	4.0	32.1	8.0	33.3	11.8
Hot source tank	326.0		326.0		326.4	
Hot sink tank	565.3		565.3		565.3	
Cold source tank	25.4		28.9		28.2	
Cold sink tank	−40.8		−40.8		−40.8	
Comp. pressure ratio	4.0005		3.9770		3.9903	
Turbine pressure ratio	3.9233		3.9066		3.9262	
Comp. efficiency	89.9931		90.0000		89.9918	
Turbine Efficiency	89.9990		90.0000		89.9971	

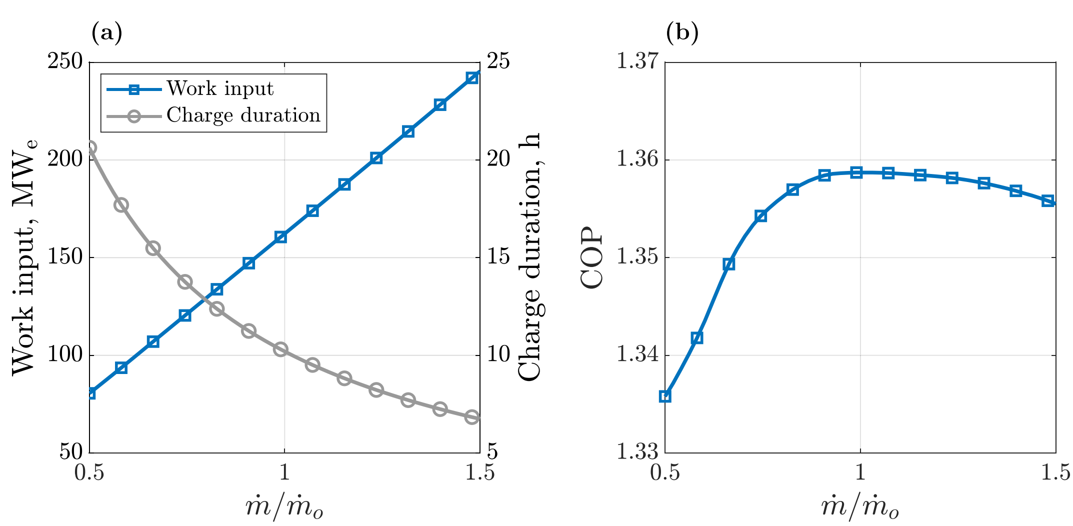

Fig. 6. (a) Power input and duration (b) COP of the charge phase as a function of the working fluid mass flow rate.

flexibility, increasing charging rates may lead to increased maintenance costs and reduce component lifetime. It should be noted that the design discharge cycle requires a larger pressure ratio than the design charge cycle: the maximum pressures in the design discharge phase are 31.4 bar – see Table 4. This suggests that it may be possible to increase the heat pump rate by 25 % while remaining within the pressure rating of the discharge phase – as long as the motor is also appropriately designed.

Fig. 6b indicates that significant changes in power input can be achieved without significant efficiency penalties, as the Coefficient of Performance remains fairly constant. By keeping the swallowing capacity constant, the pressure ratio and efficiency of the compressor and turbine remain close to their design value, as shown in Table 3. Since pressure ratios remain almost constant, temperature ratios also do, and as a result the heat exchangers deliver fluid to the sink tanks at temperatures very close to the design value. Thus, the heat pump achieves a high level of performance over a wide range of $\dot{m}_{WF}$. Furthermore, this method of off-design operation does not cause significant deviations to the sink tank temperatures (Table 3), and consequently, future discharge and charge phases will not be adversely affected by the off-design operation of this charge phase.

4.2. Heat engine mass flow rate

Varying the mass flow rate during the discharge phase affects the heat engine in a similar way to the heat pump. The heat engine power output varies linearly with $\dot{m}_{WF}$ while the discharge duration τ_{dis} is inversely proportional, as in Fig. 7a. As a result, Fig. 7b shows that the heat engine efficiency does not deviate significantly from its design value despite the large variance in $\dot{m}_{WF}$. This is due to inventory control allowing the turbomachinery to operate close to the design efficiency and pressure ratio, maintaining pressure ratios, and delivering fluid to the source tanks at close to design temperatures, as shown in Table 4. Therefore, the heat engine could conceivably be operated over a wide range of off-design points to meet the electrical load required by the grid. However, like the heat pump, this requires the system to be rated for the appropriate temperatures and generator rated for increased loads. The design discharge phase requires higher pressures than the charge phase to return the storage fluids to the correct temperatures. Therefore, increasing the heat engine power rating leads to pressures considerably higher than those in the design charge phase. However, reduced heat engine powers (for increased discharging durations) are feasible since the pressures reduce.

The results in Sections 4.1 and 4.2 are consistent with previous publications, such as Refs. [16, 19, 24], which suggest that inventory control is a powerful strategy that can allow PTES to provide load-following capabilities without significantly affecting the efficiency or storage systems.

4.3. Ambient temperature

The heat engine uses an air cooler to reject heat from the system, and as a result, ambient temperature T_{amb} variations create off-design operating conditions. Here, the design $T_{\mathrm{amb,o}}$ is 25 °C and conditions ranging from −25 °C to 65 °C are investigated. This 90 °C temperature range is larger than what may be experienced in a single practical location. For example, in Bakersfield, California, T_{amb} varied between 0.5 °C and 49.2 °C in the period 2020–2022, and in Munich, Germany it varied between −16 °C and 33.9 °C between 2017 and 2019 [40].

When T_{amb} drops below the design value, the air mass flow rate is reduced to maintain the design state points rather than over-cooling the cycle. As a result, temperatures and pressures around the cycle remain constant, as do the sink tank temperatures. The work output increases slightly because the parasitic electrical load to move air through the air cooler is reduced. Fig. 8 indicates that this can lead to a 1–2 % increase in power output. Fig. 9a demonstrates that the system pressure does not have to vary significantly to accommodate large variation in T_{amb} which is advantageous from a system design and maintenance perspective.

Ambient temperatures hotter than the design point lead to hotter working fluid temperatures leaving the air cooler ($T_{7'} > T_{7',o}$) and entering the cold heat exchanger. The control strategy prioritizes maintaining the hot tank temperatures, and as a result, the cold storage fluid is delivered to the cold source tank at temperatures higher than the design value. Fig. 9b demonstrates a linear increase in $T_{\mathrm{sink,c}}$ with T_{amb} while $T_{\mathrm{source,h}}$ remains constant. An increase in T_{amb} of 20 °C (e.g. from 25 °C to 45 °C) increases $T_{\mathrm{source,c}}$ by just under 20 °C. This excess temperature could potentially be rejected to the environment when T_{amb} decreases, but in this article, it is assumed this is not feasible.

An alternative control strategy could keep the cold source tank at the design temperature by increasing the cold storage fluid mass flow rate $\left(\dot{m}_c > \dot{m}_{c,o}\right)$. This approach is analyzed in Appendix A, which shows that the hot and cold tanks become unbalanced. Unbalanced tanks lead to lower energy capacities and rebalancing the tanks leads to an increase in cold source tank temperature. Furthermore, the overall round-trip efficiency across multiple cycles is 55.8 % which is lower than the proposed control strategy which achieves 58.8 %.

Changes to storage tank temperatures impact subsequent cycles. Results are shown for six consecutive charge-discharge cycles in Fig. 10.

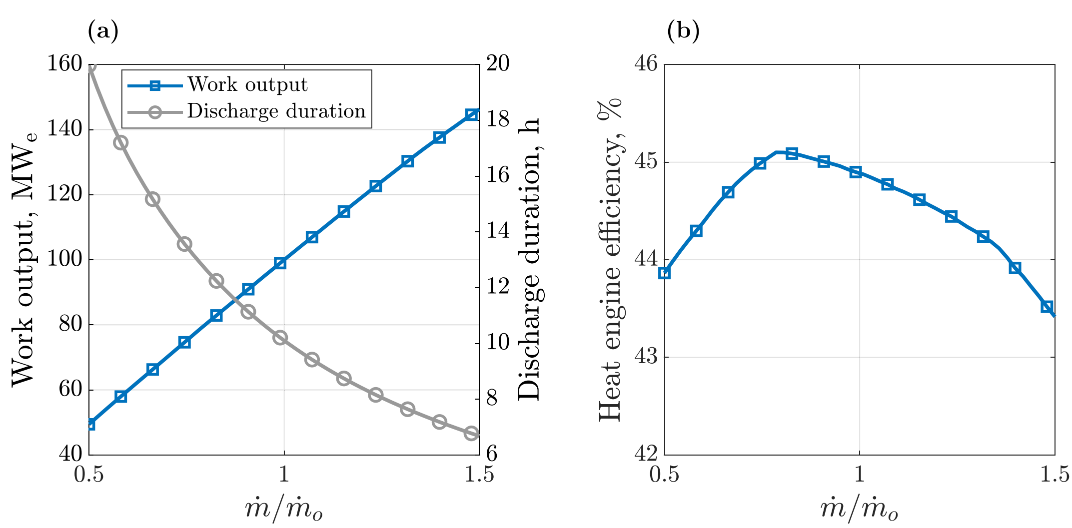

Fig. 7. (a) Power output and duration (b) Heat engine efficiency of the discharge phase as a function of the working fluid mass flow rate.

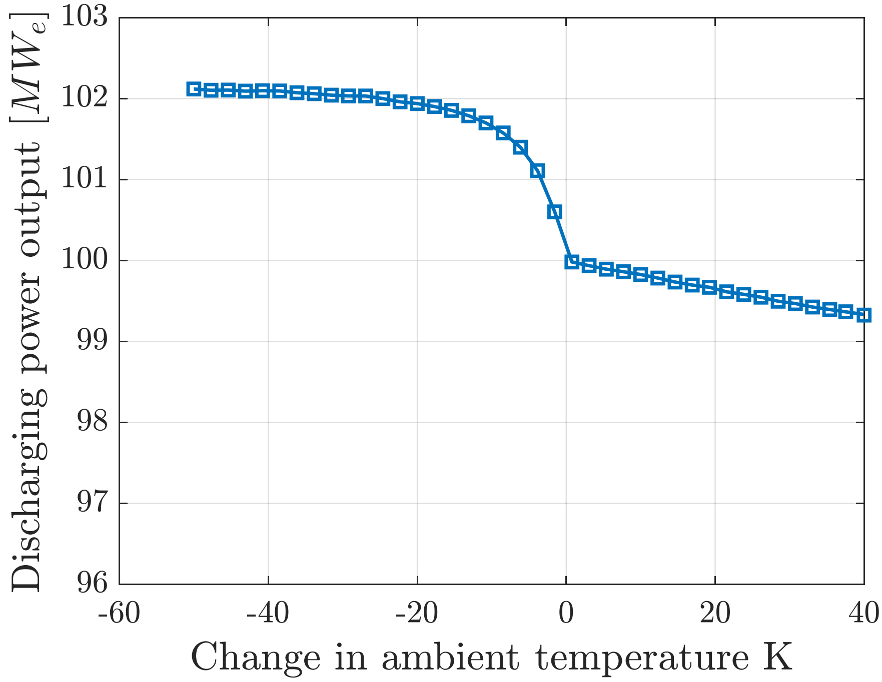

Fig. 8. Discharge power output as a function of ambient temperature.

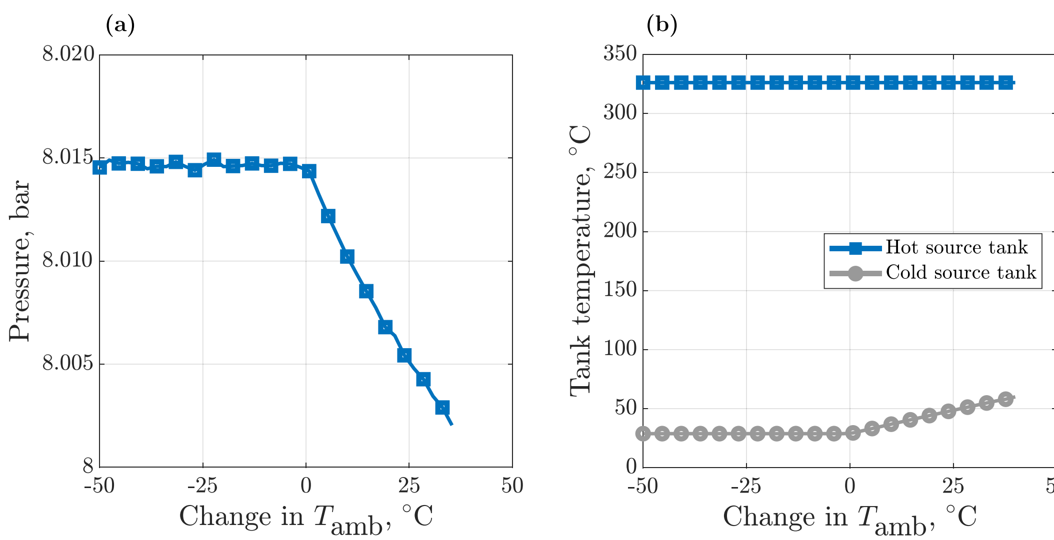

Fig. 9. (a) Variation in turbine exit pressure (point 5′) (b) Variation in hot and cold source tank temperature as a function of ambient temperature during discharge.

T_{amb} is increased by 25 °C to 50 °C for the first three cycles before returning to 25 °C for the final three cycles. T_{amb} and the temperature of the four storage tanks are illustrated for each phase in Fig. 10. Throughout the operation, the control strategy maintains the hot storage tank at the design temperature but in the first discharging phase $T_{source,c}$ increases 6 %: this impacts the next charge phase in which $T_{sink,c}$ also increases (by 7.5 %). The results demonstrate that the tank temperatures stabilize close to these values if T_{amb} remains at 50 °C. This means there is not a 'runaway' effect where tank temperatures become uncontrollable and that instead they remain acceptably close to the design values. When T_{amb} reduces in the fourth cycle, $T_{source,c}$ returns to close to its design value during the discharge phase. After two more cycles, the tank temperatures have fully 'self-corrected' to the design values. Therefore, this method of control enables the hot fluid temperature and tank SOCs to be carefully controlled while the cold fluid temperature varies by an acceptable margin.

For these results, the working fluid mass flow rate equaled the design value, and Fig. 11 shows how the power output, discharge duration, and round-trip efficiency vary for each cycle. Increased T_{amb} affects $T_{source,c}$ in the discharge phase, which then increases $T_{sink,c}$ in the following charge phase. Therefore, the impact of T_{amb} on cycle performance occurs the following cycle. Fig. 11 shows that higher cold sink temperature leads to reduced power output and efficiency in cycles 2–4 due to warmer temperatures at the compressor inlet. The discharge duration also reduces slightly. Although T_{amb} returns to normal in cycle 4, the discharge power only increases in cycle 5, and returns to the design output in cycle 6 once the tank temperatures have fully corrected.

An alternative operation strategy that harnesses the flexibility of inventory control is also shown on Fig. 11. The plant operator may decide that reduced power output is not acceptable and can increase the power output by increasing the working fluid mass flow rate as

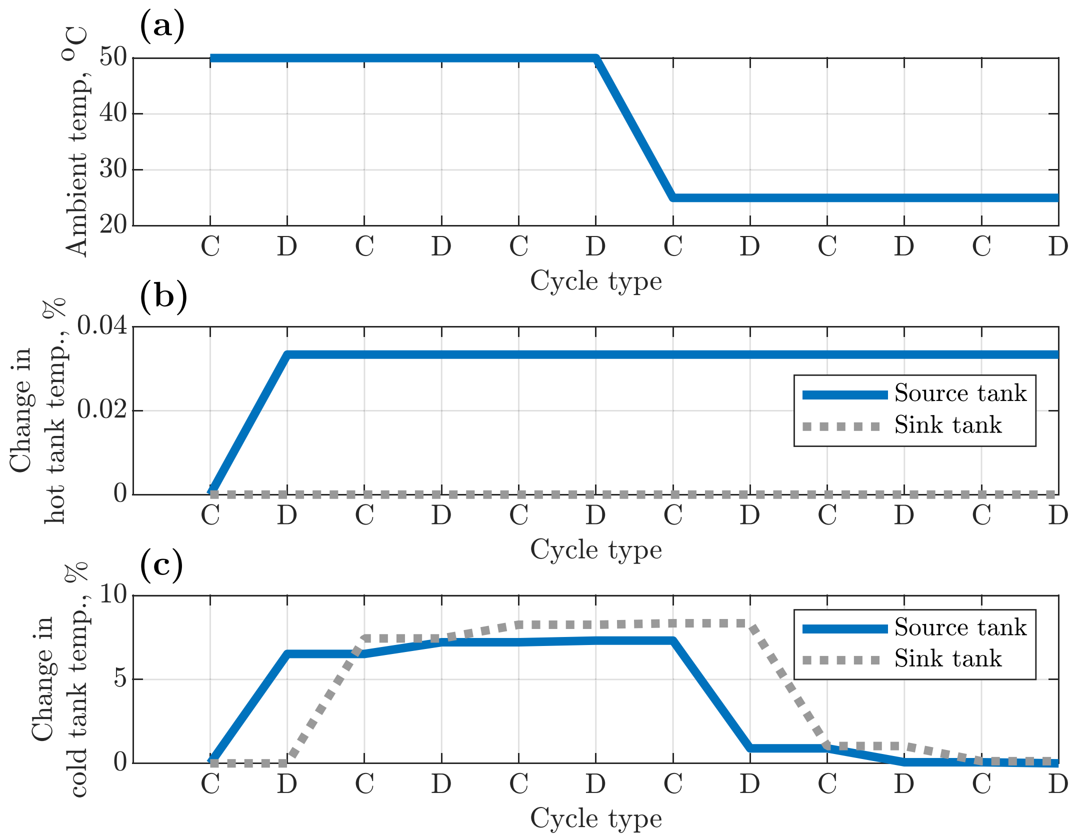

Fig. 10. Six consecutive charge-discharge cycles with high ambient temperatures for the first three cycles. (a) Ambient temperature. (b) Hot tank temperatures. (c) Cold tank temperatures.

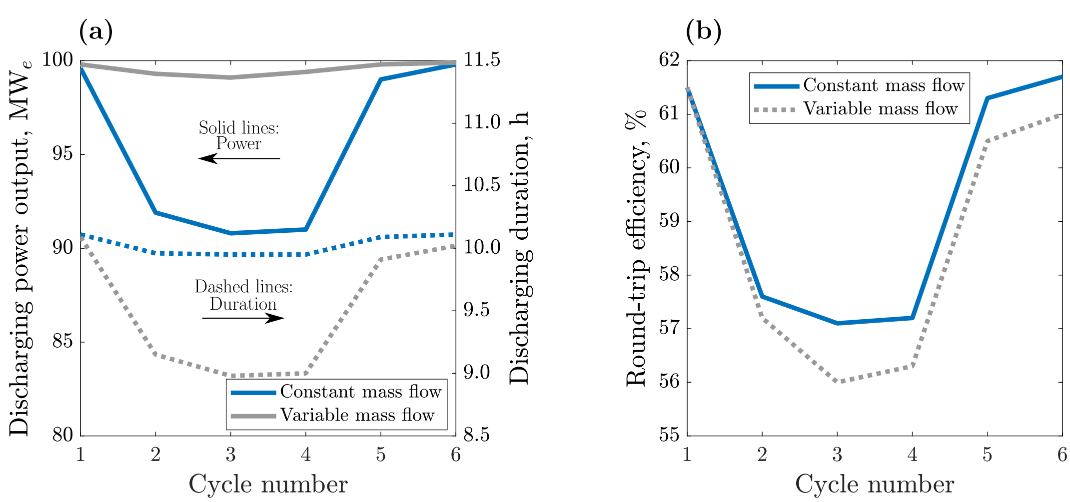

Fig. 11. Comparison of two control strategies for six consecutive charge-discharge cycles with high ambient temperatures in the first three cycles. (a) Power output and discharge duration. (b) Round-trip efficiency.

described in Section 4.2. In this example, $\dot{m}_{WF}$ must be increased by ~10 % to enable the plant to discharge at close to the design power output when T_{amb} is 25 °C greater than the design value. This corresponds to a 10 % increase in pressure which should be factored into the engineering design. Although the power output is increased, Fig. 11 demonstrates that the discharge duration decreases: this is because the same quantity of thermal energy is available for discharge at the same reduced conversion efficiency, the operator has simply chosen a different allocation between power and time. Fig. 11b indicates that varying $\dot{m}_{WF}$ to maintain power output is slightly less efficient but this may be acceptable if the plant is contracted to deliver a certain power rating.

4.4. Tank temperatures

Tank temperatures can be affected by off-design operation in previous cycles. Heat leakage into or out of the tanks during idle periods may also affect the temperature: the hot sink and source tanks are >300 °C and are likely to reduce in temperature due to heat loss, while the cold sink tank temperature is <0 °C and could increase in temperature due to heat addition. The cold source tank temperature is just above the design ambient temperature, so its temperature could increase or decrease slightly depending on ambient conditions.

Changes to the source tank temperatures T_{source} impact the charging phase heat pump and may have a subsequent impact on the sink tank temperatures T_{sink}. Fig. 12a shows how changing T_{source} of either the hot tank (blue line) or cold tank (grey line) affects $T_{\text{sink,h}}$, while Fig. 12b shows the impact on $T_{\text{sink,c}}$. If $T_{\text{source,h}}$ increases, then storage tank control seeks to maintain a constant $T_{\text{sink,h}}$ by increasing $\dot{m}_h$. Balancing the tank SOCs means that $\dot{m}_c$ increases proportionately thereby leading to increased $T_{\text{sink,c}}$. The cumulative effect of these changes is an increase in power input (Fig. 12c) and a reduction in charge duration (Fig. 12d). (Higher $T_{\text{source,h}}$ increases the working fluid temperature leaving the hot heat exchanger T_3, which means heat delivery occurs over a larger average temperature, therefore reducing the COP).

On the other hand, when $T_{\text{source,h}}$ reduces, storage fluid control reduces $\dot{m}_h$ to keep $T_{\text{sink,h}}$ close to its design point, resulting in a decreased $T_{\text{sink,c}}$. Trends in power input are somewhat more complicated and erratic due to the way in which the model seeks a converged solution. The power input generally increases as the heat pump is required to provide greater 'lift'.

Variations in $T_{\text{source,c}}$ have straightforward consequences which are also shown on Fig. 12. As $T_{\text{source,c}}$ increases, the $T_{\text{sink,h}}$ is kept constant which is achieved by increasing $\dot{m}_h$. The control strategy then increases $\dot{m}_c$, and therefore $T_{\text{sink,c}}$ increases in proportion to $T_{\text{source,c}}$, while the charge duration and power output decrease.

Changes to the sink tank temperatures T_{sink} have intuitive effects on the discharging power output, as shown in Fig. 13. Reducing $T_{\text{sink,h}}$ (or increasing $T_{\text{sink,c}}$) reduces the temperature range that the heat engine operates over, leading to reduced power output. Maintaining a constant value of $T_{\text{source,h}}$ requires $\dot{m}_h$ to increase, thereby leading to increased $\dot{m}_c$ and reduced τ_{dis}. The value of $T_{\text{source,c}}$ varies in proportion with both $T_{\text{sink,h}}$ and $T_{\text{sink,c}}$.

Fig. 12 and Fig. 13 are useful for anticipating how tank temperatures will change in the phases after an initial impulse. If the temperature of a source tank changes, then the impact on sink tanks is found on Fig. 12. These changes to the sink tanks may then affect the source tanks, as found on Fig. 13. For instance, reducing $T_{\text{sink,h}}$ leads to colder $T_{\text{source,c}}$ during discharge: in turn, in the next charge cycle, $T_{\text{sink,c}}$ is then reduced which subsequently affects the next discharge cycle. The graphs show that a small change in $T_{\text{sink,c}}$ leads to a smaller variation in $T_{\text{source,c}}$. Therefore, after several cycles, the cold tanks will return to their design values. In the meantime, the hot tank temperatures do not vary

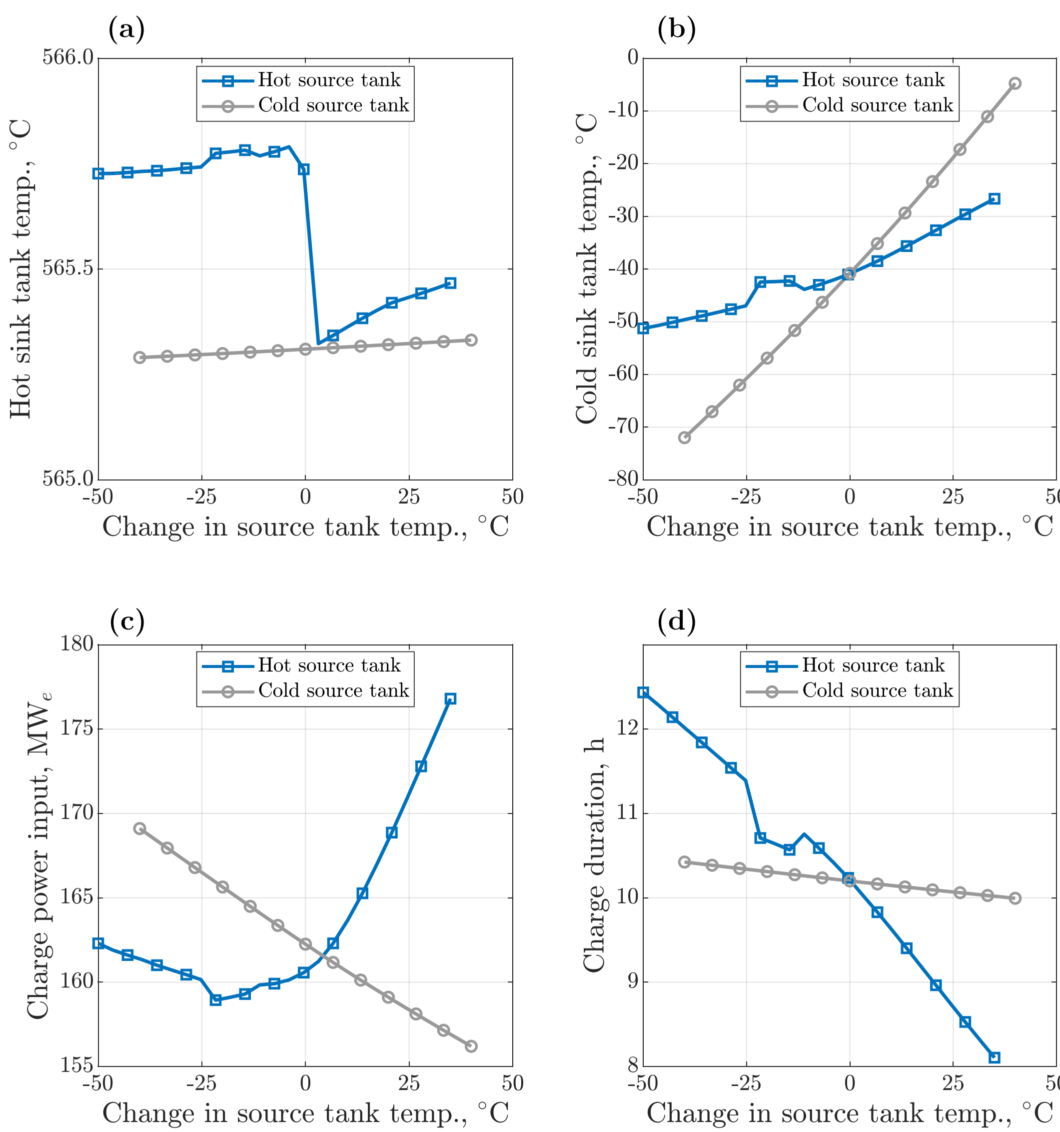

Fig. 12. Effect of variable source tank temperatures during the charge phase on (a) hot sink temperature (b) cold sink temperature (c) charge power input (d) charge duration. Results are shown for variations in both the hot source tank and cold source tank temperature.

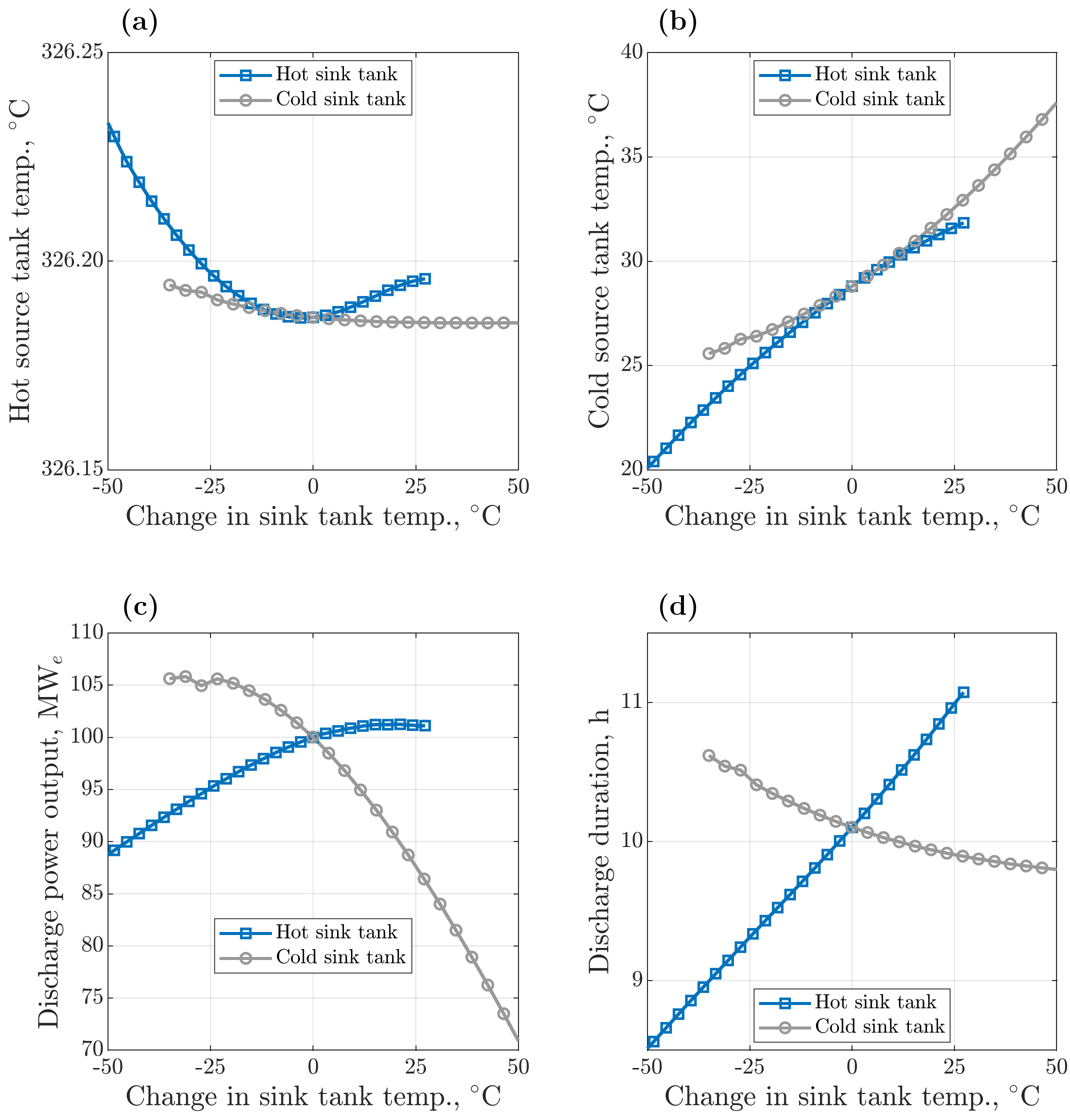

Fig. 13. Effect of variable sink tank temperatures during the discharge phase on (a) hot source temperature (b) cold source temperature (c) discharge power output (d) discharge duration. Results are shown for variations in both the hot source tank and cold source tank temperature.

significantly.

This 'self-correcting' behavior is confirmed by running several continuous charge-discharge cycles and observing the tank temperatures as shown in Fig. 14 and Table 5. In the first cycle, $T_{\mathrm{sink,h}}$ is reduced by 50 °C to represent heat leakage. This reduces the total energy stored and therefore the electricity that can be generated. Here, inventory control is implemented to ensure that the discharge power output is maintained close to the design value of 100 MW$_{\mathrm{e}}$ despite the reduced storage capacity. These results demonstrate that inventory control can deliver the required power even when off-design conditions exist. However, the discharge duration and round-trip efficiency of the first charge-discharge cycle are reduced compared to the nominal values. Furthermore, the discharge mass flow rate must be increased by 11 %, and the system pressure increases correspondingly. Thus, the maximum pressure in discharge increases from 31.5 bar to 34.5 bar. Plant designers must therefore consider the range of likely off-design scenarios and the desired control when designing maximum pressure limits.

The control of the storage fluid mass flow rates manages the tank temperatures and SOCs. Results show that $T_{\mathrm{source,h}}$ does not vary but that $T_{\mathrm{source,c}}$ reduces. This affects the second cycle in which $T_{\mathrm{sink,c}}$ is reduced by 7 °C. Although this increases the heat pump work input, it increases the heat engine energy output (which operates over a larger temperature difference) and as a result the round-trip efficiency of the second cycle is slightly improved. Cold tank temperatures begin to return towards their design values in the third cycle, and as a result the duration and round-trip efficiency also quickly converge on the nominal values.

5. Conclusions

Pumped Thermal Energy Storage (PTES) is an electricity storage device suitable for long-duration energy storage, and this article focuses on a Joule-Brayton PTES system with liquid thermal energy storage. Much work has been devoted to developing and improving system designs, and recent work has started to focus on understanding how to operate these systems in off-design scenarios. Off-design conditions arise for several reasons: they can be intentional if the system is providing load-following capabilities in which case the working fluid mass flow rate is varied to deliver the required power input or output. Alternatively, ambient temperature variations or temperature changes in the storage tanks (due to heat leakage, for example) also move the system away from its design point.

Understanding off-design behavior and developing robust control strategies is important because such operation can reduce the performance of the current cycle and subsequent cycles. For instance, we show

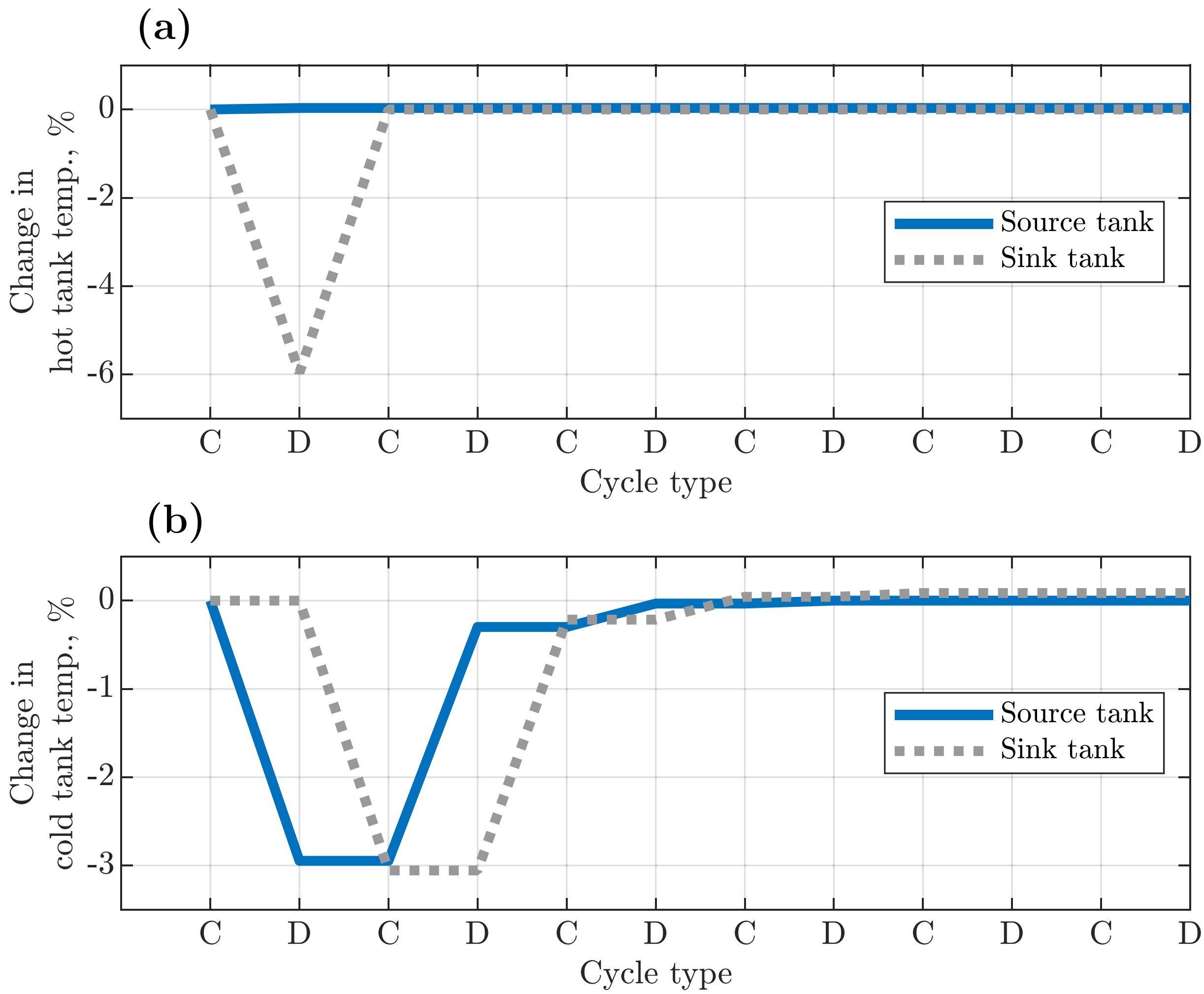

Fig. 14. Six consecutive charge-discharge cycles with an unexpected decrease in hot sink temperature in the first discharge phase. (a) Hot tank temperatures. (b) Cold tank temperatures.

Table 5
Results for six consecutive cycles. In the first cycle, the hot sink tank temperature is reduced. Tank temperatures are shown at the end of the charge-discharge cycle.

Cycle #	Work in, MW_e	Work out, MW_e	Charge duration, h	Discharge duration, h	Round-trip efficiency, %	$T_{source,h}$, °C	$T_{sink,h}$, °C	$T_{source,c}$, °C	$T_{sink,c}$, °C
1	160.4	99.1	10.2	7.6	46.1	326.3	515.3	20.3	−40.8
2	162.3	100	10.2	10.4	62.7	326.2	565.3	28.1	−47.7
3	161.0	99.9	10.2	10.1	61.7	326.2	565.3	28.8	−41.3
4	160.9	99.9	10.2	10.1	61.5	326.2	565.3	28.9	−40.6
5	160.9	99.9	10.2	10.1	61.5	326.2	565.3	28.9	−40.6
6	160.9	99.9	10.2	10.1	61.5	326.2	565.3	28.9	−40.6

that changes in ambient temperature can affect the temperature of the cold storage tank, which therefore impacts the performance of the next cycle.

In this article, an off-design model and control strategy for a Joule-Brayton PTES system with liquid thermal energy storage is developed. The control strategy has two components (1) Inventory control is used to manage the power input/output of the closed thermodynamic cycle and ensure steady-state cyclic behavior (2) Control of the storage fluids mass flow rate to ensure the storage tanks have the correct temperatures and state-of-charge (SOC). There are many ways to implement these strategies and the plant operator can prioritize different aspects of system performance, such as power output, discharge duration, acceptable variations in tank temperatures, or allowing the tanks to have different SOC.

We propose a strategy that prioritizes temperature management of the hot storage fluid, which in this case is molten salt, which must be kept from freezing in the cooler tank, and from decomposing in the hotter tank. The hot fluid temperature also has a more significant impact on round-trip efficiency and power output than the cold fluid. The hot fluid mass flow rate is therefore varied in off-design scenarios to maintain the hot fluid as close to the design temperatures as possible. On the other hand, the cold fluid mass flow rate is varied to ensure that the cold tanks have the same SOC as the hot tanks. This leads to the cold fluid being stored at temperatures that differ from the design point. However, this is deemed acceptable because the cold fluid (methanol) has a more flexible operating temperature range than the hot fluid, and temperature changes are observed to be small. Variation of the cold tank temperatures impacts subsequent cycles. For instance, we find that increased ambient temperatures lead to increased temperatures in the cold tank, and this leads to reduced energy output in the next charge-discharge cycle. Importantly, the control strategy leads to stable behavior: when ambient temperatures are increased 100 %, we show that after three charge-discharge cycles, the cold tanks stabilize at a new temperature 6 % higher than the design value. When the ambient temperature returns to normal, the cold tanks quickly return to their design temperatures after three cycles. We show that this approach is preferable to controlling the system to maintain cold tank design temperatures, as this leads to the hot tanks having a different SOC to the cold tank, and management of this condition leads to a lower round-trip efficiency when considering multiple charge-discharge cycles.

We also investigate the impact of changes to the storage tank temperatures. Heat leakage from the hot tanks (or heat addition to the cold tanks) effectively reduces the energy stored, which therefore decreases the energy available in discharge – this materializes as either a reduced power output, reduced discharge duration, or both. We show that changes in tank temperatures affect several subsequent charge-discharge cycles: hot tank temperatures are maintained close to the design point, but the cold tank temperatures vary to a small extent. However, after two charge-discharge cycles all the tanks return to their design values which demonstrates the stable nature of the control strategy.

We demonstrate the capabilities of inventory control which is used to allow changes to the mass flow rate in the thermodynamic cycles. The working fluid mass flow rate can be varied to deliver a target power output or input, although this leads to a trade-off with the phase duration. This feature can be used to provide load-following capabilities at the design point. Furthermore, we show that this approach can be used to manage unexpected off-design conditions. For instance, increased ambient temperatures or heat leakage from the hot tanks can lead to reduced power outputs if the working fluid mass flow rate is kept constant. By using inventory control, the power output can be restored to its design value despite the off-design conditions that occur.

There are several avenues for future work. This article introduced the off-design control strategy and applied it to simple scenarios. Subsequent analysis should investigate the impact of more complicated scenarios where multiple parameters have off-design values. The analysis here considered multiple charge-discharge cycles in response to a single off-design change – such as elevated ambient temperature. More realistic load cycles should be investigated where off-design conditions change continuously. Longer time horizons should be considered – such as the performance over the course of a year. Such work would help quantify the actual round-trip efficiency that PTES can achieve under real conditions. Furthermore, the potential of off-design operation and inventory control to maximize the value of PTES to the grid should be investigated. Economic decisions are likely to affect the system operation and may outweigh any impacts on efficiency. We showed that inventory control allowed the required power input/output to be delivered during off-design conditions. But further analysis is required to understand what economically optimal load profiles may look like and whether PTES can meet those requirements. Finally, off-design operation leads to variations in the maximum system pressures and power ratings, and therefore these models can be used to inform system design, such as pressure limits and power ratings of turbomachinery and ancillary equipment.

CRediT authorship contribution statement

Joshua McTigue: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Ty Neises:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was authored by the National Renewable Energy Laboratory (NREL), operated by Alliance for Sustainable Energy, LLC, for the U.S. Department of Energy (DOE) under Contract No. DE-AC36-08GO28308. Funding provided by U.S. Department of Energy: Office of Energy Efficiency and Renewable Energy, Solar Energy Technologies Office. The views expressed in the article do not necessarily represent the views of the DOE or the U.S. Government. The U.S. Government retains and the publisher, by accepting the article for publication, acknowledges that the U.S. Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for U.S. Government purposes.

Appendix A. Alternative control strategy

The storage fluid control strategy aims to keep the hot storage tanks at the design temperature, while the cold fluid mass flow rate varies to ensure the hot and cold tanks have equal state-of-charge (SOC). As a result, the temperature of the cold fluid can deviate from its design value. As an example, Table 6 illustrates how tank temperatures and SOCs vary over six charge-discharge cycles when the first three cycles have elevated ambient temperatures – these results correspond to those shown in Fig. 11. The ambient temperatures are 25 °C greater than the design value, and this leads to the cold source and sink tank temperatures increasing roughly 20 °C. However, the tanks are operated to their maximum extents – i.e. the SOC varies between 0 % and 100 % each cycle, and the hot tanks and cold tanks have equal SOCs.

On the other hand, the plant operator may wish to avoid cold tank temperature variations. Table 7 illustrates a scenario where the cold storage fluid mass flow rate is varied to maintain the cold fluid at the design temperature. This is achieved by increasing $\dot{m}_c$, and this results in the cold sink tank being fully discharged before the hot sink tank – see row D1, which shows that only 73 % of the hot fluid is returned to the hot source tank. As a result, in the next charging phase (C2) there is less hot fluid available to be heated up (and moved from the source tank to the sink tank). Therefore, the hot source tank is depleted before the cold source tank is fully emptied, and this reduces the charge duration considerably. This pattern continues while ambient temperatures remain high with charge and discharge phases becoming shorter, less effectively utilizing the tanks and reducing the energy stored.

The ambient temperature returns to the design value in the final three charge-discharge cycles, but the hot and cold tanks have very different SOCs. The tanks can be rebalanced by adjusting the cold fluid mass flow rate: in the fourth charge-discharge cycle, the hot and cold tanks are both returned to equal SOC, which enables them to be fully utilized. However, this does lead to a large increase in the cold storage fluid temperature. Therefore, this control strategy does not manage to completely avoid variations in cold tank temperatures, and unbalancing the storage tanks leads to less energy being stored (7.6 GWh_e compared to 9.7 GWh_e) and a lower average round-trip efficiency (55.8 % compared to 58.8 %) than the strategy shown in

Table 6. Overall, this method of operation is less effective than the one employed throughout the article.

Table 6
Detailed results for six charge-discharge cycles with off-design ambient temperature. Storage fluid control strategy maintains equal SOC for hot and cold tanks. SOCs and tank temperatures are shown at the end of each charge/discharge phase.

Cycle #	T_{amb}, °C	Work, MW_e	Duration, h	Hot source, SOC, %	Cold source SOC, %	$T_{source,h}$, °C	$T_{sink,h}$, °C	$T_{source,c}$, °C	$T_{sink,c}$, °C
C1	50	−160.4	10.2	0	0	326.0	565.3	28.9	−40.8
D1	50	99.8	10.1	100	100	326.2	565.3	48.6	−40.8
C2	50	−160.4	9.9	0	0	326.2	565.3	48.6	−23.4
D2	50	99.3	9.2	100	100	326.2	565.3	50.5	−23.4
C3	50	−160.5	9.9	0	0	326.2	565.3	50.5	21.4
D3	50	99.1	9.0	100	100	326.2	565.3	50.6	−21.4
C4	25	−160.5	9.9	0	0	326.2	565.3	50.6	−21.3
D4	25	99.4	9.0	100	100	326.2	565.3	31.3	−21.3
C5	25	−160.4	10.2	0	0	326.2	565.3	31.3	−38.2
D5	25	99.8	9.9	100	100	326.2	565.3	28.6	−38.2
C6	25	−160.4	10.2	0	0	326.2	565.3	28.6	−40.5
D6	25	99.8	10.0	100	100	326.2	565.3	28.2	−40.5

Table 7
Detailed results for six charge-discharge cycles with off-design ambient temperature. The storage fluid control strategy keeps the storage fluids at the design temperature in the first three cycles. The final three cycles seek to restore equal SOC in the hot and cold tanks.

Cycle #	T_{amb}, °C	Work, MW_e	Duration, h	Hot source, SOC, %	Cold source SOC, %	$T_{source,h}$, °C	$T_{sink,h}$, °C	$T_{source,c}$, °C	$T_{sink,c}$, °C
C1	50	−160.3	10.2	0	0	326.0	565.3	28.9	−41.2
D1	50	99.6	7.5	73.1	100	326.2	565.3	29.1	−41.2
C2	50	−158.6	7.5	0	28.5	326.2	565.3	29.1	−40.6
D2	50	99.6	5.4	53.1	100	326.2	565.3	29.3	−40.6
C3	50	−159.9	5.4	0	50.1	326.2	565.3	29.3	−42.3
D3	50	99.6	3.9	37.3	100	326.2	565.3	29.4	−42.3
C4	25	−161.3	3.9	0	0	326.2	565.3	29.4	4.7
D4	25	97.6	7.2	100	100	326.2	565.3	35.6	4.7
C5	25	−160.4	10.3	0	0	326.2	565.3	35.6	−30.9
D5	25	99.6	9.3	100	100	326.2	565.3	26.7	−30.9
C6	25	−160.4	10.5	0	0	326.2	565.3	26.7	−38.3
D6	25	99.8	9.7	100	100	326.2	565.3	25.3	−38.3

References

[1] P. Denholm, W. Cole, N. Blair, Moving Beyond 4-Hour Li-Ion Batteries: Challenges and Opportunities for Long(er)-Duration Energy Storage, NREL Tech. Rep. - NREL/TP-6A40-85878. https://www.nrel.gov/docs/fy23osti/85878.pdf, 2023.

[2] Department for Energy Security& Net Zero, Scenario Deployment Analysis for Long-Duration Electricity Storage: A study of the benefits of Long-Duration Electricity Storage technologies on the GB power system, DESNZ Res. Pap. Number 2023/047. https://assets.publishing.service.gov.uk/media/659be546c23a1000128d0c51/long-duration-electricity-storage-scenario-deployment-analysis.pdf, 2024.

[3] International Energy Agency, Net Zero by 2050: A Roadmap for the Global Energy Sector, iea.li/nzeroadmap, 2021.

[4] California Public Utilities Commission, Decision Determining Need for Centralized Procurement of Long Lead-Time Resources. https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M536/K273/536273174.PDF, 2024.

[5] C. Augustine, N. Blair, Storage Technology Modeling Input Data Report Storage Technology Modeling Input Data Report, NREL Tech. Rep. - NREL/TP-5700-78694. https://www.nrel.gov/docs/fy21osti/78694.pdf, 2021.

[6] A.V. Olympios, J.D. McTigue, P. Sapin, C.N. Markides, Pumped-Thermal Electricity Storage Based on Brayton Cycles, Elsevier Inc., 2021, https://doi.org/10.1016/b978-0-12-819723-3.00086-x.

[7] A. Vecchi, K. Knobloch, T. Liang, H. Kildahl, A. Sciacovelli, K. Engelbrecht, Y. Li, Y. Ding, Carnot battery development: a review on system performance, applications and commercial state-of-the-art, J. Energy Storage 55 (2022) 105782, https://doi.org/10.1016/j.est.2022.105782.

[8] International Energy Agency (IEA), Task 36 Carnot Batteries Final Report. https://iea-es.org/publications/final-report-task-36/, 2024.

[9] M.T. Ameen, Z. Ma, A. Smallbone, R. Norman, A.P. Roskilly, Demonstration system of pumped heat energy storage (PHES) and its round-trip efficiency, Appl. Energy 333 (2023) 120580, https://doi.org/10.1016/j.apenergy.2022.120580.

[10] J. Howes, Concept and development of a pumped heat electricity storage device, Proc. IEEE 100 (2012) 493–503, https://doi.org/10.1109/JPROC.2011.2174529.

[11] A.J. White, J.D. McTigue, C.N. Markides, Analysis and optimisation of packed-bed thermal reservoirs for electricity storage applications, Proc. Inst. Mech. Eng. Part A J. Power Energy 230 (2016) 739–754.

[12] T.J. Held, Low-cost, long-duration electrical energy storage using a CO2-based Electro Thermal Energy Storage (ETES) system, in: ARPA-E DAYS Annual Meeting, 2021.

[13] T.J. Held, J. Miller, J. Mallinak, V. Avadhanula, L. Magyar, Pilot-scale testing of a trans- critical CO2-based Pumpted Thermal Energy Storage (PTES) system, Proc. ASME Turbo Expo 2024 (2024).

[14] Office of Clean Energy Demonstrations, Pumped thermal energy stOrage in ALaska Railbelt (POLAR). https://www.energy.gov/oced/long-duration-energy-storage-demonstrations-projects-selected-and-awarded-projects#polar, 2024 (accessed July 26, 2024).

[15] N.R. Smith, J. Just, J. Johnson, F.K. Bulnes, Performance characterization of a small-scale pumped thermal energy storage system, in: Proc. ASME Turbo Expo, Boston, MA, 2023, pp. 1–12.

[16] J.D. McTigue, P. Farres-Antunez, C.N. Markides, A.J. White, Pumped thermal energy storage with liquid storage, Encycl. Energy Storage (2021), https://doi.org/10.1016/B978-0-12-819723-3.00054-8.

[17] H.U. Frutschi, Closed-Cycle Gas Turbines: Operating Experience and Future Potential, 1st ed., ASME, 2005.

[18] T. Neises, Steady-state off-design modeling of the supercritical carbon dioxide recom- pression cycle for concentrating solar power applications with two-tank sensible-heat storage, Sol. Energy 212 (2020) 19–33, https://doi.org/10.1016/j.solener.2020.10.041.

[19] G.F. Frate, L. Paternostro, L. Ferrari, U. Desideri, Off-design of a Pumped Thermal Energy Storage Based on Closed Brayton Cycles, in: Proc. ASME Turbo Expo 2-21, 2021, pp. 1–14.

[20] J.D. McTigue, C.N. Markides, A.J. White, Performance response of packed-bed thermal storage to cycle duration perturbations, J. Energy Storage 19 (2018) 379–392, https://doi.org/10.1016/j.est.2018.08.016.

[21] L. Wang, X. Lin, L. Chai, L. Peng, D. Yu, H. Chen, Cyclic transient behavior of the Joule–Brayton based pumped heat electricity storage: modeling and analysis, Renew. Sust. Energ. Rev. 111 (2019) 523–534, https://doi.org/10.1016/j.rser.2019.03.056.

[22] A. Albay, Z. Zhu, M. Mercangöz, State-of-Charge (SoC) Management of PTES Coupled Industrial Cogeneration Systems, in: Proc. ASME Turbo Expo 2024, London, UK, 2024.

[23] A. Benato, M. Pecchini, F. De Vanna, A. Stoppato, Off-design analysis of a TES-based electricity storage system, Proc. ECOS. (2023) 2277–2287.

[24] G.F. Frate, M. Pettinari, E. Di Pino Incognito, R. Costanzi, L. Ferrari, Dynamic modelling of a Brayton Ptes system, Proc. ASME Turbo Expo. 4 (2022) 1–14, https://doi.org/10.1115/GT2022-83445.

[25] C. Lu, X. Shi, Q. He, Y. Liu, X. An, S. Cui, D. Du, Dynamic modeling and numerical investigation of novel pumped thermal electricity storage system during startup process, J. Energy Storage 55 (2022) 105409, https://doi.org/10.1016/j.est.2022.105409.

[26] H. Yang, J. Li, Z. Ge, L. Yang, X. Du, Dynamic characteristics and control strategy of pumped thermal electricity storage with reversible Brayton cycle, Renew. Energy (2022), https://doi.org/10.1016/j.renene.2022.08.129.

[27] H. Yang, J. Li, Z. Ge, L. Yang, X. Du, Dynamic performance for discharging process of pumped thermal electricity storage with reversible Brayton cycle, Energy 263 (2023) 125930, https://doi.org/10.1016/j.energy.2022.125930.

[28] M. Pettinari, G.F. Frate, A.P. Tran, J. Oehler, P. Stathopoulos, L. Ferrari, Transient analysis and control of a Brayton heat pump during start-up, in: 36th Int. Conf. Effic. Cost, Optim. Simul. Environ. Impact Energy Syst. ECOS 2023, 2023, pp. 839–850, https://doi.org/10.52202/069564-0076.

[29] M. Pettinari, G.F. Frate, K. Kyprianidis, L. Ferrari, Dynamic Modelling and Part-Load Behavior of a Brayton Heat Pump, Proc. 64th Int. Conf. Scand. Simul. Soc. SIMS 2023 Västerås, Sweden, Sept. 25–28, 2023 200, 2023, pp. 254–261, https://doi.org/10.3384/ecp200033.

[30] G.F. Frate, A. Baccioli, L. Bernardini, L. Ferrari, Assessment of the off-design performance of a solar thermally-integrated pumped-thermal energy storage, Renew. Energy 201 (2022) 636–650, https://doi.org/10.1016/j.renene.2022.10.097.

[31] P. Wang, Q. Li, S. Wang, C. He, C. Wu, Off-design performance evaluation of thermally integrated pumped thermal electricity storage systems with solar energy, Energy Convers. Manag. 301 (2024) 118001, https://doi.org/10.1016/j.enconman.2023.118001.

[32] M. Mehos, C. Turchi, J. Vidal, M. Wagner, Z. Ma, C. Ho, W. Kolb, C. Andraka, A. Kruizenga, Concentrating Solar Power Gen3 Demonstration Roadmap, NREL Tech. Report, NREL/TP-5500-67464, 2017.

[33] J.D. McTigue, P. Farres-Antunez, K. Sundarnath J, C.N. Markides, A.J. White, Techno-economic analysis of recuperated Joule-Brayton pumped thermal energy storage, Energy Convers. Manag. 252 (2022), https://doi.org/10.1016/j.enconman.2021.115016.

[34] I.H. Bell, J. Wronski, S. Quoilin, V. Lemort, Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp, Ind. Eng. Chem. Res. 53 (2014) 2498–2508, https://doi.org/10.1021/ie4033999.

[35] A.S. Nagorny, N.V. Dravid, R.H. Jansen, B.H. Kenny, Design aspects of a high speed permanent magnet synchronous motor/generator for flywheel applications, in: 2005 IEEE Int. Conf. Electr. Mach. Drives, 2005, pp. 635–641, https://doi.org/10.1109/iemdc.2005.195790.

[36] F. Incropera, D. DeWitt, T. Bergman, A. Lavine, Fundamentals of Heat and Mass Transfer, 7th Edition, John Wiley and Sons, 2011.

[37] N. Zhang, R. Cai, Analytical solutions and typical characteristics of part-load performan- ces of single shaft gas turbine and its cogeneration, Energy Convers. Manag. 43 (2002) 1323–1337, https://doi.org/10.1016/S0196-8904(02)00018-3.

[38] A. Sciacovelli, Y. Li, H. Chen, Y. Wu, J. Wang, S. Garvey, Y. Ding, Dynamic simulation of Adiabatic Compressed Air Energy Storage (A-CAES) plant with integrated thermal storage – link between components performance and plant performance, Appl. Energy 185 (2017) 16–28, https://doi.org/10.1016/j.apenergy.2016.10.058.

[39] J. Martinek, J. Jorgenson, J.D. McTigue, On the operational characteristics and economic value of pumped thermal energy storage, J. Energy Storage 52 (2022) 105005, https://doi.org/10.1016/j.est.2022.105005.

[40] National Solar Radiation Database, (n.d.). https://nsrdb.nrel.gov/ (accessed November 12, 2017).